

Complex Numbers 1, group *Alpha* α -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{239} + 4i^{353} - 6i^{-683} + 4$??? -3

(b) $7x^2 + 4x + 4 = 0$??? $-\frac{2}{7} \pm \frac{\sqrt{208}}{14}i$

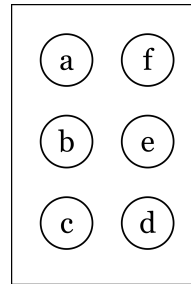
(c) $(1 - 3i)(x + 3yi) = -1 + 5i$.. ??? .. $-23/15$

(d) $2(-8 - 5i) - (3 - 3i)\overline{(5 + 4i)}$??? $-19 - 7i$

(e) $\left(\frac{-1+2i}{-3+3i}\right)^2$??? $5/18$

(f) $(e^{i2\pi})^{-5} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^5$... ??? ... $e^{i\frac{4\pi}{3}}$

1.



Braille letter

Complex Numbers 1, group *Alpha* α -ii

Name:

- (a) Determine the imaginary part.
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 (d) Write in the algebraic form.
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 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{288} + 9i^{557} - i^{-400} - 3$??? 9

(b) $5x^2 - x + 6 = 0$??? $\frac{1}{10} \pm \frac{\sqrt{239}}{10}i$

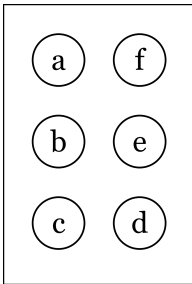
(c) $(-3 - 2i)(-3x + yi) = -1 - 4i$. ??? . $-53/39$

(d) $-2(7 + 6i) - (-2 + 4i)\overline{(1 + 5i)}$??? $-52 - 26i$

(e) $\left(\frac{-1-i}{-2+3i}\right)^2$??? $-24/169$

(f) $(e^{i\frac{2\pi}{3}})^{-3} \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)^9$??? $e^{i\frac{3\pi}{2}}$

2.



Braille letter

Complex Numbers 1, group *Alpha* α -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{260} - 6i^{383} - 9i^{-939} - 1$??? -3

(b) $3x^2 + 5x + 7 = 0$??? $-\frac{5}{6} \pm \frac{\sqrt{59}}{3}i$

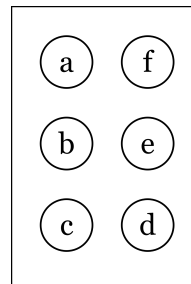
(c) $(-3 + 8i)(x + yi) = 1 - 4i$??? $-55/73$

(d) $4(-6 + 2i) - (2 + 5i)\overline{(-7 + 2i)}$??? $-30 + 47i$

(e) $\left(\frac{3+i}{2+3i}\right)^2$??? $32/169$

(f) $(e^{i\frac{4\pi}{3}})^{-3} (\cos \pi + i \sin \pi)^{-8}$??? $1e^{i\pi}$

3.



Braille letter

Complex Numbers 1, group *Alpha* α -iv

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 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{799} + i^{533} - 6i^{-110} + 9$??? -4

(b) $2x^2 - 4x + 5 = 0$??? $1 \pm \frac{\sqrt{24}}{4}i$

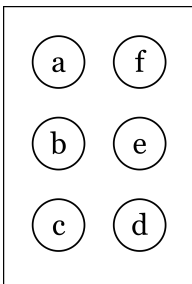
(c) $(3 + 2i)(-x - 6yi) = 5 - 2i$. ??? . $-25/39$

(d) $1(4 - 5i) - (-8 - 3i)\overline{(1 + 3i)}$??? $5 - 26i$

(e) $\left(\frac{1-i}{2-2i}\right)^2$??? $1/4$

(f) $(-2e^{i\frac{5\pi}{4}})^5 (\cos \pi + i \sin \pi)^{-5}$??? $-32e^{i\frac{5\pi}{4}}$

4.



Braille letter

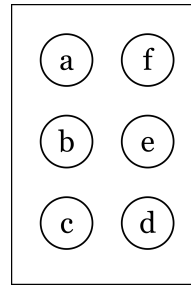
Complex Numbers 1, group *Beta* β -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
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 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{564} + 6i^{731} + i^{-850} + 6$??? -6
 (b) $3x^2 - x + 1 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{11}}{6}i$
 (c) $(-2 - i)(x + 7yi) = -2 - i$??? 1
 (d) $-6(6 + 8i) - (-4 + 2i)\overline{(5 + 3i)}$??? -62 - 70i
 (e) $\left(\frac{-3+i}{-1-2i}\right)^2$??? -16/3
 (f) $(3e^{i\pi})^5 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)^{-7}$... ??? ... $243e^{i\frac{\pi}{4}}$

1.



Braille letter

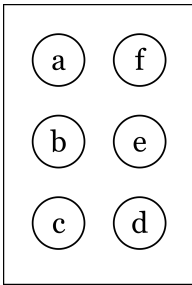
Complex Numbers 1, group *Beta* β -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{302} - 4i^{38} + 4i^{-443} - 2$??? 4
 (b) $5x^2 - 6x + 2 = 0$??? $\frac{3}{5} \pm \frac{2}{5}i$
 (c) $(1 + i)(4x - 4yi) = 6 + 4i$??? 0
 (d) $2(3 - 6i) - (-7 + 5i)\overline{(-2 - 4i)}$??? 32 + 26i
 (e) $\left(\frac{7+4i}{1+i}\right)^2$??? $65/2$
 (f) $(e^{i2\pi})^{-3} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)^{-5}$... ??? ... $e^{i\frac{5\pi}{4}}$

2.



Braille letter

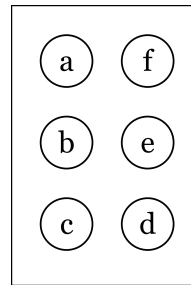
Complex Numbers 1, group *Beta* β -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{807} + 3i^{50} - 4i^{-815} + 9$??? -5
 (b) $x^2 + 4x + 5 = 0$??? $-2 \pm 2i$
 (c) $(3 - 4i)(-6x - yi) = 2 + 4i$... ??? ... -11/15
 (d) $-5(7 - 7i) - (-3 + 3i)\overline{(8 - 8i)}$??? -35 + 35i
 (e) $\left(\frac{3+3i}{1-2i}\right)^2$??? -72/25
 (f) $(-4e^{i\frac{2\pi}{3}})^2 \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}\right)^{-3}$.. ??? .. $16e^{i\frac{2\pi}{15}}$

3.



Braille letter

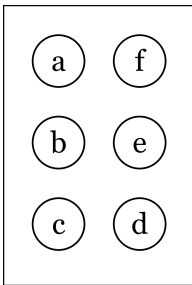
Complex Numbers 1, group *Beta* β -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{207} + 2i^{353} - 4i^{-959} + 5$??? -3
 (b) $8x^2 + x + 6 = 0$??? $\frac{-1}{16} \pm \frac{\sqrt{191}}{8}i$
 (c) $(-1 + 2i)(-3x - yi) = -4 + 3i$??? -43/15
 (d) $1(7 - 3i) - (9 - 4i)\overline{(-1 - i)}$??? -6 - 16i
 (e) $\left(\frac{-4-3i}{1-2i}\right)^2$??? -117/25
 (f) $(6e^{i\pi})^2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{-9}$.. ??? .. $36e^{i\pi}$

4.



Braille letter

Complex Numbers 1, group *Gamma* γ -i

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- (a) Determine the imaginary part.
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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{103} - i^{540} + 6i^{-913} + 6$??? -7

(b) $4x^2 + 6x + 3 = 0$??? $\frac{-3}{4} \pm \frac{\sqrt{60}}{8}i$

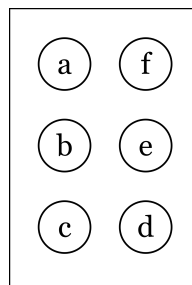
(c) $(-2 + 4i)(-3x - yi) = 3 - 3i$??? $-3/5$

(d) $1(-9 + 2i) - (-5 + 2i)\overline{(-4 - 4i)}$??? -21 - 10i

(e) $\left(\frac{-3+3i}{1+2i}\right)^2$??? $-18/5$

(f) $\left(e^{i\frac{3\pi}{2}}\right)^{-4} (\cos 2\pi + i \sin 2\pi)^3$??? 1

1.



Braille letter

Complex Numbers 1, group *Gamma* γ -ii

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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{486} - 3i^{901} + 2i^{-498} - 1$??? -3

(b) $5x^2 + 5x + 7 = 0$??? $\frac{-1}{2} \pm \frac{\sqrt{115}}{5}i$

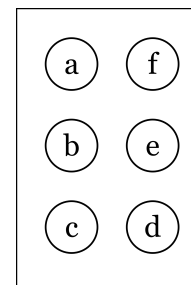
(c) $(3 + 2i)(3x + 3yi) = -2 - 7i$??? $-3/13$

(d) $-4(-8 + i) - (-1 - 5i)\overline{(-2 - 5i)}$??? -20 - 9i

(e) $\left(\frac{-2+5i}{1-2i}\right)^2$??? $29/5$

(f) $\left(e^{i\pi}\right)^{-4} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^6$??? $e^{i\frac{13\pi}{12}}$

2.



Braille letter

Complex Numbers 1, group *Gamma* γ -iii

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 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{660} + i^{629} + 4i^{-249} - 4$??? 6

(b) $7x^2 - 4x + 4 = 0$??? $\frac{2}{7} \pm \frac{\sqrt{96}}{14}i$

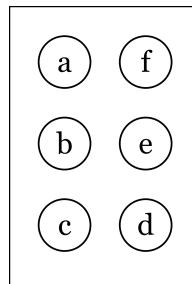
(c) $(2 + 4i)(-x - 4yi) = 2 + i$??? $-13/40$

(d) $-5(1 - 9i) - (3 + 7i)\overline{(-3 + 3i)}$??? -35 + 75i

(e) $\left(\frac{-8+8i}{-1+i}\right)^2$??? 64

(f) $\left(e^{i\frac{9\pi}{7}}\right)^{-3} \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)^{-2}$... ??? ... $e^{i\frac{17\pi}{21}}$

3.



Braille letter

Complex Numbers 1, group *Gamma* γ -iv

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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{610} + i^{639} + 8i^{-546} - 1$??? -1

(b) $4x^2 - 3x + 2 = 0$??? $\frac{3}{8} \pm \frac{\sqrt{23}}{8}i$

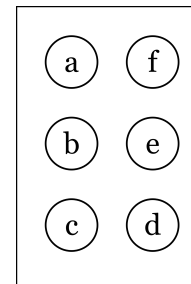
(c) $(-2 - i)(8x + yi) = 5 + 2i$??? $-21/10$

(d) $3(-7 - 4i) - (-2 + 2i)\overline{(8 - 2i)}$??? -33 - 24i

(e) $\left(\frac{2-2i}{-1-i}\right)^2$??? -4

(f) $\left(e^{i\pi}\right)^{-2} \left(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}\right)^3$??? $e^{i\frac{9\pi}{5}}$

4.



Braille letter

Complex Numbers 1, group *Delta* δ -i

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- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{65} + 7i^{423} + 4i^{-859} - 8$??? -2

(b) $8x^2 + 2x + 7 = 0$??? $-\frac{1}{8} \pm \frac{\sqrt{220}}{16}i$

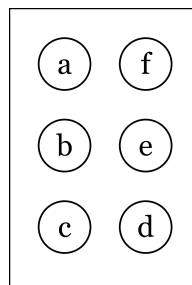
(c) $(3 + i)(2x + 5yi) = -1 + 3i$??? $1/5$

(d) $5(-1 + 3i) - (2 + 2i)\overline{(6 + 4i)}$??? -25 + 11i

(e) $\left(\frac{-1+i}{-3+3i}\right)^2$??? $1/9$

(f) $\left(e^{i\frac{3\pi}{2}}\right)^{-7} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-3}$... ??? ... $1e^{i\pi}$

1.



Braille letter

Complex Numbers 1, group *Delta* δ -ii

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- (a) Determine the imaginary part.
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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{394} - 5i^{518} - i^{-90} - 6$??? -5

(b) $x^2 + x + 5 = 0$??? $-\frac{1}{2} \pm \frac{\sqrt{19}}{2}i$

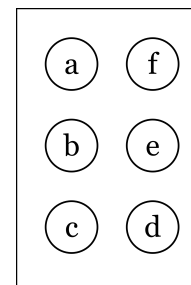
(c) $(4 - i)(-2x + yi) = -2 - i$... ??? ... $-3/34$

(d) $5(-1 - 4i) - (-2 - i)\overline{(2 + 8i)}$??? -1 - 34i

(e) $\left(\frac{-2-5i}{1-2i}\right)^2$??? -17

(f) $\left(e^{i\frac{3\pi}{2}}\right)^{-2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{-5}$.. ??? .. $e^{i\frac{5\pi}{3}}$

2.



Braille letter

Complex Numbers 1, group *Delta* δ -iii

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- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{37} + i^{883} + 2i^{-747} - 5$??? 2

(b) $7x^2 - 6x + 6 = 0$??? $\frac{3}{7} \pm \frac{\sqrt{132}}{7}i$

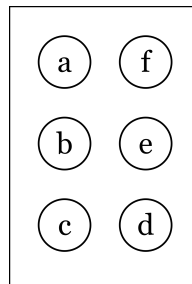
(c) $(1 + i)(2x + 7yi) = -7 - 6i$... ??? ... $-65/28$

(d) $-3(-1 - 5i) - (1 - 3i)\overline{(-1 - i)}$??? -2 + 11i

(e) $\left(\frac{2-8i}{1+i}\right)^2$??? -16

(f) $\left(e^{i2\pi}\right)^{-5} \left(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7}\right)^3$??? $e^{i\frac{11\pi}{7}}$

3.



Braille letter

Complex Numbers 1, group *Delta* δ -iv

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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{558} + 7i^{364} - 5i^{-729} - 8$??? 5

(b) $2x^2 - 2x + 4 = 0$??? $\frac{1}{2} \pm \frac{\sqrt{28}}{4}i$

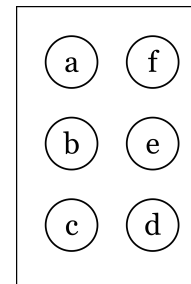
(c) $(-2 + 4i)(-x - 6yi) = 5 - i$.. ??? .. $17/20$

(d) $7(8 - 3i) - (-1 - 3i)\overline{(-2 + i)}$??? 57 - 28i

(e) $\left(\frac{1-4i}{-2-2i}\right)^2$??? -1

(f) $\left(e^{i\frac{2\pi}{3}}\right)^{-4} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^{-6}$... ??? ... $e^{i\frac{13\pi}{36}}$

4.



Braille letter

Complex Numbers 1, group *Epsilon* € -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{108} - i^{952} - 6i^{-826} - 7$??? 0

(b) $3x^2 - 2x + 6 = 0$??? $\frac{1}{3} \pm \frac{\sqrt{68}}{6}i$

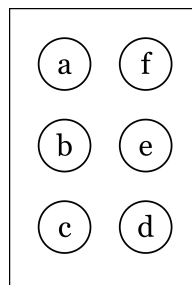
(c) $(-3 - 3i)(x + 5yi) = 6 + 5i$.. ??? .. $-22/15$

(d) $-4(6 + 5i) - (2 + 8i)\overline{(1 + i)}$??? $-34 - 30i$

(e) $\left(\frac{-6+3i}{-1-i}\right)^2$??? -18

(f) $\left(e^{i\frac{5\pi}{7}}\right)^{-1} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-4}$.. ??? .. $e^{i\frac{19\pi}{14}}$

1.



Braille letter

Complex Numbers 1, group *Epsilon* € -ii

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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{516} + 4i^{899} + 9i^{-403} + 3$??? 5

(b) $4x^2 - 3x + 3 = 0$??? $\frac{3}{8} \pm \frac{\sqrt{39}}{4}i$

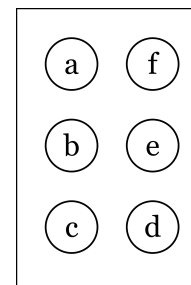
(c) $(3 + i)(x + 2yi) = -5 + i$??? -1

(d) $-5(-8 - 5i) - (-4 + 5i)\overline{(2 + 6i)}$??? $-12 - 9i$

(e) $\left(\frac{-2+3i}{1+i}\right)^2$??? -6

(f) $\left(e^{i\frac{\pi}{6}}\right)^{-3} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)^6$??? $1e^{i\pi}$

2.



Braille letter

Complex Numbers 1, group *Epsilon* € -iii

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 (d) Write in the algebraic form.
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 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{664} - 9i^{163} - 2i^{-280} + 3$??? 9

(b) $8x^2 + 4x + 5 = 0$??? $-\frac{1}{4} \pm \frac{3}{4}i$

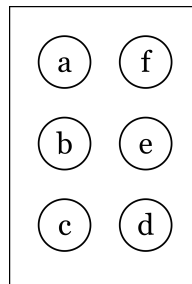
(c) $(-3 - i)(-3x + 3yi) = 4 + 4i$... ??? ... $4/15$

(d) $1(-1 + i) - (-4 - 4i)\overline{(-1 + 3i)}$??? $15 - 15i$

(e) $\left(\frac{-4+2i}{1-i}\right)^2$??? -10

(f) $\left(e^{i\pi}\right)^{-5} \left(\cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9}\right)^{-3}$??? $e^{i\frac{\pi}{3}}$

3.



Braille letter

Complex Numbers 1, group *Epsilon* € -iv

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 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{154} + 5i^{38} + 5i^{-842} - 4$??? 0

(b) $4x^2 + 2x + 8 = 0$??? $-\frac{1}{4} \pm \frac{\sqrt{252}}{8}i$

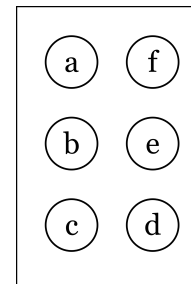
(c) $(1 - 3i)(-2x - 2yi) = 3 + 2i$.. ??? .. $-1/5$

(d) $-4(1 - 2i) - (2 - 6i)\overline{(4 + 5i)}$??? $48 + 42i$

(e) $\left(\frac{2+2i}{1-3i}\right)^2$??? $-12/25$

(f) $\left(e^{i\frac{\pi}{4}}\right)^{-1} \left(\cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}\right)^{-1}$.. ??? .. $e^{i\frac{19\pi}{20}}$

4.



Braille letter

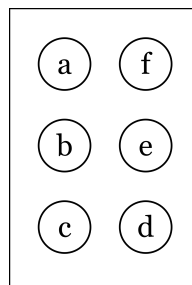
Complex Numbers 1, group *Zeta* ζ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{69} + 6i^{607} + i^{-985} + 5$??? -6
 (b) $2x^2 - 3x + 3 = 0$??? $\frac{3}{4} \pm \frac{\sqrt{15}}{4}i$
 (c) $(2 + 5i)(-4x - yi) = 4 - i$.. ??? .. $75/116$
 (d) $7(-6 + i) - (-7 - 2i)\overline{(1 - i)}$??? -37 + 2i
 (e) $\left(\frac{2+i}{-3-3i}\right)^2$??? 2/9
 (f) $(-9e^{i\frac{3\pi}{2}})^2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)^3$.. ??? .. 81

1.



Braille letter

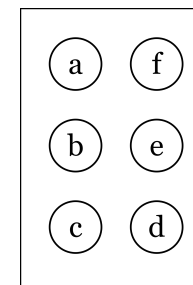
Complex Numbers 1, group *Zeta* ζ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{150} - i^{250} + 3i^{-227} + 2$??? 3
 (b) $2x^2 + x + 3 = 0$??? $-\frac{1}{4} \pm \frac{\sqrt{23}}{4}i$
 (c) $(5 + 2i)(x - yi) = -3 + 2i$??? $-27/29$
 (d) $5(-4 - 3i) - (-5 + 8i)\overline{(-7 + 2i)}$??? -87 + 31i
 (e) $\left(\frac{3-i}{1-2i}\right)^2$??? 0
 (f) $(e^{i2\pi})^{-7} \left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8}\right)^{-2}$??? $e^{i\frac{3\pi}{4}}$

2.



Braille letter

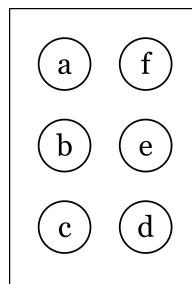
Complex Numbers 1, group *Zeta* ζ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{686} - 3i^{118} + 5i^{-392} - 7$??? 0
 (b) $8x^2 + 7x + 3 = 0$??? $-\frac{7}{16} \pm \frac{\sqrt{143}}{16}i$
 (c) $(-2 - 3i)(-2x + 5yi) = 4 + 4i$. ??? . $14/13$
 (d) $-2(1 - 5i) - (-7 - i)\overline{(1 + 6i)}$??? 17 - 31i
 (e) $\left(\frac{-2+i}{1+3i}\right)^2$??? -1/2
 (f) $(3e^{i\pi})^5 \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}\right)^2$.. ??? .. $243e^{i\frac{3\pi}{7}}$

3.



Braille letter

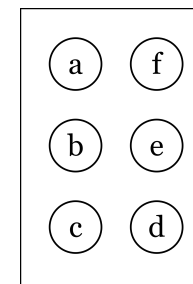
Complex Numbers 1, group *Zeta* ζ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{959} - 2i^{683} - 4i^{-848} + 6$??? 1
 (b) $x^2 - x + 7 = 0$??? $\frac{1}{2} \pm \frac{\sqrt{55}}{2}i$
 (c) $(-7 + 2i)(x + yi) = 1 - 5i$??? $16/53$
 (d) $-7(-4 + 4i) - (-6 + i)\overline{(-4 + 4i)}$??? 0 - 48i
 (e) $\left(\frac{-5-3i}{1+2i}\right)^2$??? $72/25$
 (f) $(e^{i\frac{\pi}{5}})^{-2} \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)^5$??? $e^{i\frac{11\pi}{10}}$

4.



Braille letter

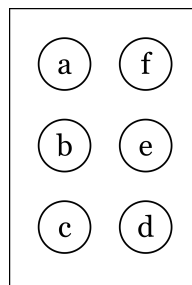
Complex Numbers 1, group *Eta* η -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{342} + 2i^{277} + 2i^{-190} - 1$??? 5
 (b) $7x^2 - x + 2 = 0$??? $\frac{1}{14} \pm \frac{\sqrt{55}}{14}i$
 (c) $(-3 - 5i)(x + yi) = -2 + i$.. ??? .. $-6/17$
 (d) $-2(3 - 3i) - (1 + 5i)\overline{(-7 - i)}$??? $11 + 40i$
 (e) $\left(\frac{2+i}{1-2i}\right)^2$??? 1
 (f) $(-2e^{i\frac{7\pi}{4}})^2 (\cos \pi + i \sin \pi)^{-4}$.. ??? .. $4e^{i\frac{3\pi}{2}}$

1.



Braille letter

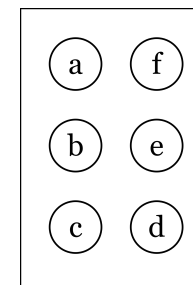
Complex Numbers 1, group *Eta* η -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{729} + 2i^{648} + 3i^{-738} - 2$??? 1
 (b) $2x^2 + x + 4 = 0$??? $\frac{-1}{4} \pm \frac{\sqrt{63}}{4}i$
 (c) $(5 + 7i)(x - yi) = -7 + i$... ??? ... $8/37$
 (d) $5(2 - 5i) - (1 + i)\overline{(1 - 2i)}$??? $11 - 24i$
 (e) $\left(\frac{-7-2i}{-1+i}\right)^2$??? -14
 (f) $(-4e^{i\frac{5\pi}{7}})^3 (\cos \pi + i \sin \pi)^5$??? $-64e^{i\frac{17\pi}{14}}$

2.



Braille letter

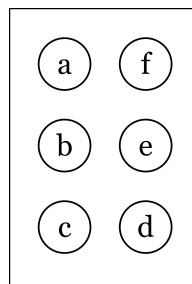
Complex Numbers 1, group *Eta* η -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{792} - 6i^{963} + 6i^{-615} - 3$??? 12
 (b) $x^2 - x + 1 = 0$??? $\frac{1}{2} \pm \sqrt{3}i$
 (c) $(3 + i)(-5x - yi) = 1 - 4i$??? $-27/25$
 (d) $-7(1 + 2i) - (-2 - 2i)\overline{(-2 + 5i)}$??? $-1 - 8i$
 (e) $\left(\frac{-1+4i}{2+i}\right)^2$??? $-77/9$
 (f) $(e^{i\frac{2\pi}{7}})^{-5} (\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})^{-6}$??? $e^{i\frac{62\pi}{35}}$

3.



Braille letter

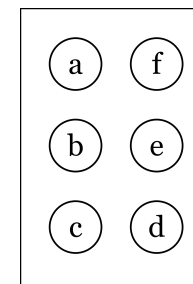
Complex Numbers 1, group *Eta* η -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{770} - 2i^{804} + 3i^{-654} + 4$??? 2
 (b) $8x^2 - 3x + 7 = 0$??? $\frac{3}{16} \pm \frac{\sqrt{215}}{16}i$
 (c) $(-7 - 3i)(-x - 2yi) = -1 - 3i$??? $-25/58$
 (d) $-4(-1 - 3i) - (6 + i)\overline{(2 + 2i)}$??? $14 + 22i$
 (e) $\left(\frac{1-3i}{-2-2i}\right)^2$??? $-3/4$
 (f) $(e^{i\frac{\pi}{2}})^{-1} (\cos 2\pi + i \sin 2\pi)^3$... ??? ... $e^{i\frac{3\pi}{2}}$

4.



Braille letter

Complex Numbers 1, group *Theta* θ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{454} - 7i^{805} - 2i^{-475} + 3 \dots \dots \dots ??? \dots \dots -8$

(b) $8x^2 + 2x + 1 = 0 \dots \dots \dots ??? \dots \dots -\frac{1}{8} \pm \frac{\sqrt{28}}{16}i$

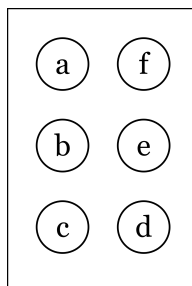
(c) $(3 - i)(2x + yi) = -1 - 6i \dots \dots \dots ??? \dots \dots -7/4$

(d) $-2(-5 - i) - (-3 + 3i)\overline{(7 - i)} \dots \dots \dots ??? \dots \dots 37 - 16i$

(e) $\left(\frac{2-5i}{1-i}\right)^2 \dots \dots \dots ??? \dots \dots \dots 10$

(f) $\left(e^{i\frac{\pi}{4}}\right)^{-2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)^5 \dots \dots \dots ??? \dots \dots e^{i\frac{\pi}{4}}$

1.



Braille letter

Complex Numbers 1, group *Theta* θ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{388} - i^{536} - 2i^{-94} - 6 \dots \dots \dots ??? \dots \dots -2$

(b) $7x^2 + 4x + 4 = 0 \dots \dots \dots ??? \dots \dots -\frac{2}{7} \pm \frac{\sqrt{96}}{14}i$

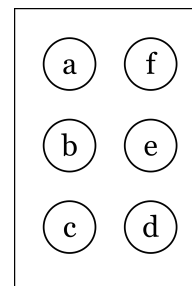
(c) $(-4 + 4i)(x - 4yi) = -2 - i \dots \dots \dots ??? \dots \dots 5/32$

(d) $-7(-8 + 4i) - (2 + i)\overline{(3 + 8i)} \dots \dots \dots ??? \dots \dots 42 - 47i$

(e) $\left(\frac{3-i}{-2+3i}\right)^2 \dots \dots \dots ??? \dots \dots \dots 32$

(f) $\left(e^{i\frac{4\pi}{3}}\right)^{-1} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)^6 \dots \dots \dots ??? \dots \dots e^{i\frac{\pi}{6}}$

2.



Braille letter

Complex Numbers 1, group *Theta* θ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{243} + i^{760} - 3i^{-143} - 2 \dots \dots \dots ??? \dots \dots -4$

(b) $x^2 - 2x + 2 = 0 \dots \dots \dots ??? \dots \dots 1 \pm \frac{\sqrt{12}}{2}i$

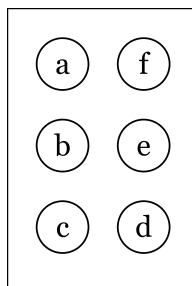
(c) $(-2 + 3i)(-5x - 2yi) = 5 + 4i \dots \dots \dots ??? \dots \dots 111/130$

(d) $-3(-1 + 8i) - (-2 + i)\overline{(-8 - 7i)} \dots \dots \dots ??? \dots \dots 1 - 2i$

(e) $\left(\frac{5+i}{2-2i}\right)^2 \dots \dots \dots ??? \dots \dots \dots -13/4$

(f) $\left(-3e^{i\frac{\pi}{2}}\right)^6 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)^{-1} \dots \dots \dots ??? \dots \dots 729e^{i\frac{3\pi}{2}}$

3.



Braille letter

Complex Numbers 1, group *Theta* θ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{488} + i^{629} + 7i^{-339} + 2 \dots \dots \dots ??? \dots \dots 8$

(b) $x^2 - 3x + 3 = 0 \dots \dots \dots ??? \dots \dots \frac{3}{2} \pm \sqrt{3}i$

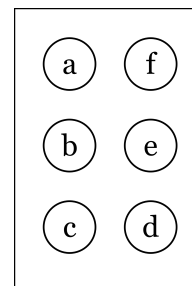
(c) $(-5 + 3i)(-3x + yi) = 1 - 2i \dots \dots \dots ??? \dots \dots 25/51$

(d) $5(-2 - i) - (-1 + i)\overline{(-3 + 4i)} \dots \dots \dots ??? \dots \dots -17 + 2i$

(e) $\left(\frac{-4+4i}{1+i}\right)^2 \dots \dots \dots ??? \dots \dots \dots -16$

(f) $\left(e^{i\frac{2\pi}{9}}\right)^{-1} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{-6} \dots \dots \dots ??? \dots \dots e^{i\frac{49\pi}{27}}$

4.



Braille letter

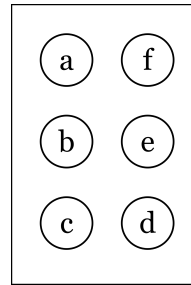
Complex Numbers 1, group *Iota* ι -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{705} + 5i^{156} - 3i^{-771} - 4$??? 3
 (b) $4x^2 - 8x + 7 = 0$??? $1 \pm \frac{\sqrt{48}}{8}i$
 (c) $(-1 - 3i)(-2x - 3yi) = -4 + i$... ??? ... $23/60$
 (d) $-4(-7 + 3i) - (-3 - 3i)\overline{(-5 + 2i)}$??? $19 - 21i$
 (e) $\left(\frac{3+4i}{-1-2i}\right)^2$??? $117/25$
 (f) $\left(e^{i\frac{8\pi}{5}}\right)^{-3} \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)^{-2}$??? $e^{i\frac{8\pi}{15}}$

1.



Braille letter

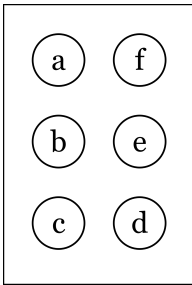
Complex Numbers 1, group *Iota* ι -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{422} + 6i^{435} - 2i^{-881} + 6$??? -4
 (b) $2x^2 - x + 3 = 0$??? $\frac{1}{4} \pm \frac{\sqrt{23}}{4}i$
 (c) $(-1 + 2i)(-3x + 2yi) = -2 - 6i$... ??? ... $5/3$
 (d) $-1(-8 + 5i) - (-4 - i)\overline{(4 - 4i)}$??? $-12 + 15i$
 (e) $\left(\frac{3+3i}{2+i}\right)^2$??? $72/25$
 (f) $\left(e^{i\frac{3\pi}{4}}\right)^{-5} \left(\cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}\right)^5$??? $e^{i\frac{15\pi}{28}}$

2.



Braille letter

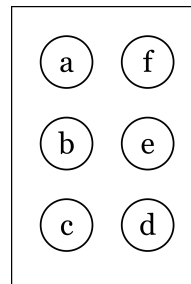
Complex Numbers 1, group *Iota* ι -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{431} + 7i^{267} + i^{-901} - 3$??? 9
 (b) $4x^2 - x + 3 = 0$??? $\frac{1}{8} \pm \frac{\sqrt{47}}{8}i$
 (c) $(-1 + 4i)(-2x - 2yi) = 2 + 6i$??? $-12/17$
 (d) $-3(6 - 5i) - (6 - i)\overline{(-3 + 7i)}$??? $14 + 54i$
 (e) $\left(\frac{-3+3i}{-1-2i}\right)^2$??? -8
 (f) $\left(e^{i\frac{5\pi}{7}}\right)^{-3} (\cos \pi + i \sin \pi)^5$??? $e^{i\frac{6\pi}{7}}$

3.



Braille letter

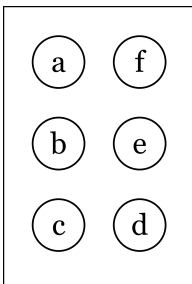
Complex Numbers 1, group *Iota* ι -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{379} + 2i^{192} - 9i^{-578} - 1$??? -1
 (b) $9x^2 + 5x + 3 = 0$??? $\frac{-5}{18} \pm \frac{\sqrt{83}}{18}i$
 (c) $(-1 + i)(4x + 2yi) = 7 + i$.. ??? .. $-11/4$
 (d) $4(-3 - 5i) - (3 - 5i)\overline{(4 + 5i)}$??? $26 + 15i$
 (e) $\left(\frac{1-i}{-3-3i}\right)^2$??? $-1/9$
 (f) $\left(7e^{i2\pi}\right)^3 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)^2$.. ??? .. $343e^{i\frac{\pi}{2}}$

4.



Braille letter

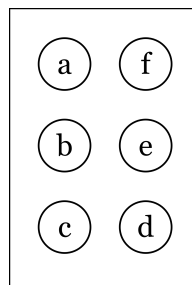
Complex Numbers 1, group *Kappa* κ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{850} - 2i^{642} - 2i^{-847} + 8$??? -3
 (b) $7x^2 + x + 2 = 0$??? $\frac{-1}{14} \pm \frac{\sqrt{55}}{14}i$
 (c) $(2 + 3i)(-4x - yi) = -9 - 2i$??? $-20/13$
 (d) $8(-4 + i) - (2 + 2i)(\overline{5 - 2i})$??? $-38 + 2i$
 (e) $\left(\frac{-1-4i}{-2+i}\right)^2$??? $-17/5$
 (f) $(e^{i\pi})^{-7} \left(\cos \frac{7\pi}{9} + i \sin \frac{7\pi}{9}\right)^7$... ??? ... $e^{i\frac{4\pi}{9}}$

1.



Braille letter

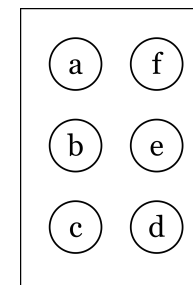
Complex Numbers 1, group *Kappa* κ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{330} - 2i^{956} + 2i^{-805} - 4$??? -2
 (b) $3x^2 - 3x + 5 = 0$??? $\frac{1}{2} \pm \frac{\sqrt{51}}{3}i$
 (c) $(4 - 2i)(x + 5yi) = -3 - 3i$... ??? ... $-9/25$
 (d) $-4(-1 - 2i) - (-8 - 8i)(\overline{2 + i})$??? $-4 + 16i$
 (e) $\left(\frac{4+3i}{-2-i}\right)^2$??? $117/25$
 (f) $(-e^{i\frac{\pi}{3}})^9 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)^5$... ??? ... $-e^{i\frac{5\pi}{3}}$

2.



Braille letter

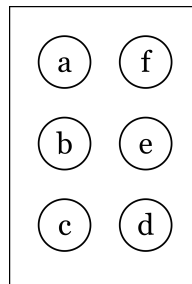
Complex Numbers 1, group *Kappa* κ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{999} - 7i^{927} - i^{-334} - 4$??? 6
 (b) $4x^2 + 4x + 3 = 0$??? $\frac{-1}{2} \pm \frac{\sqrt{32}}{4}i$
 (c) $(3 + 3i)(-2x + 2yi) = -7 - 7i$??? 0
 (d) $-4(-5 - 5i) - (-3 + 4i)(\overline{8 + i})$??? $-8 - 15i$
 (e) $\left(\frac{3+4i}{2+i}\right)^2$??? 3
 (f) $(e^{i2\pi})^{-4} (\cos 2\pi + i \sin 2\pi)^5$??? $1e^{i\pi}$

3.



Braille letter

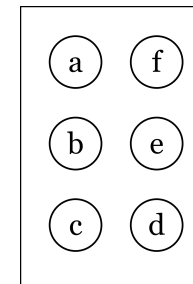
Complex Numbers 1, group *Kappa* κ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{234} + i^{499} - 7i^{-678} - 4$??? -1
 (b) $x^2 + 2x + 2 = 0$??? $-1 \pm 2i$
 (c) $(1 - i)(-2x + 6yi) = 5 - 2i$.. ??? .. $-7/6$
 (d) $-3(-7 - i) - (1 - 3i)(\overline{6 - i})$??? $9 + 20i$
 (e) $\left(\frac{-3+5i}{-1-2i}\right)^2$??? $34/5$
 (f) $(5e^{i\frac{2\pi}{7}})^3 \left(\cos \frac{9\pi}{7} + i \sin \frac{9\pi}{7}\right)^4$??? $125e^{i\pi}$

4.



Braille letter

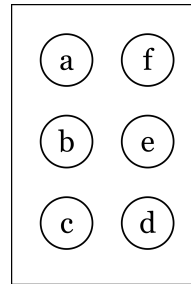
Complex Numbers 1, group *Lambda* λ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{717} - i^{746} - 7i^{-131} - 2$??? -6
 (b) $3x^2 - 2x + 4 = 0$??? $\frac{1}{3} \pm \frac{\sqrt{44}}{6}i$
 (c) $(2 + i)(6x - 2yi) = 3 - 3i$??? $6/5$
 (d) $-4(-2 + 5i) - (-3 - 3i)\overline{(-2 - 3i)}$??? $5 - 17i$
 (e) $\left(\frac{2-3i}{1-2i}\right)^2$??? $-13/5$
 (f) $(e^{i\pi})^{-3} (\cos \pi + i \sin \pi)^3$??? 1

1.



Braille letter

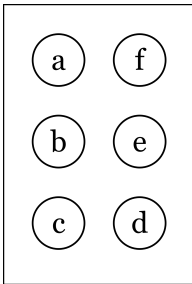
Complex Numbers 1, group *Lambda* λ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{50} + 8i^{426} - 2i^{-939} + 1$??? -2
 (b) $2x^2 - 6x + 7 = 0$??? $\frac{3}{2} \pm \frac{\sqrt{76}}{4}i$
 (c) $(-1 + 3i)(-x - 2yi) = -1 - 7i$. ??? . $-27/10$
 (d) $-3(2 + 2i) - (-1 - 4i)\overline{(-6 + 2i)}$??? $4 - 32i$
 (e) $\left(\frac{-3-2i}{-2+i}\right)^2$??? $-33/25$
 (f) $(e^{i\frac{\pi}{5}})^{-3} (\cos \pi + i \sin \pi)^3$??? $e^{i\frac{11\pi}{25}}$

2.



Braille letter

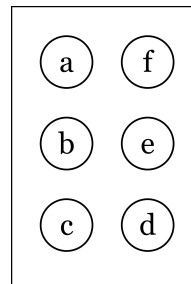
Complex Numbers 1, group *Lambda* λ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{518} - 4i^{74} + 7i^{-523} - 7$??? 7
 (b) $3x^2 - 2x + 1 = 0$??? $\frac{1}{3} \pm \frac{\sqrt{20}}{6}i$
 (c) $(-2 + 6i)(x - 2yi) = -3 - i$... ??? ... $1/5$
 (d) $-2(-5 + 2i) - (5 - 4i)\overline{(-6 + 5i)}$??? $0 - 3i$
 (e) $\left(\frac{-4+3i}{-2+2i}\right)^2$??? 3
 (f) $(e^{i\frac{5\pi}{6}})^{-5} (\cos \pi + i \sin \pi)^{-1}$... ??? ... $e^{i\frac{13\pi}{15}}$

3.



Braille letter

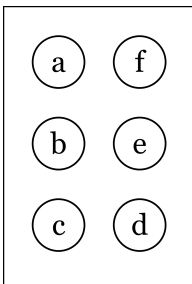
Complex Numbers 1, group *Lambda* λ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{264} + 7i^{580} + 4i^{-895} + 5$??? 12
 (b) $4x^2 + 3x + 6 = 0$??? $\frac{-3}{8} \pm \frac{\sqrt{87}}{8}i$
 (c) $(-3 + 3i)(x - 4yi) = 3 - i$.. ??? .. $-7/12$
 (d) $6(3 + 6i) - (5 - 3i)\overline{(2 - 7i)}$??? $-13 + 77i$
 (e) $\left(\frac{4+4i}{1+i}\right)^2$??? 16
 (f) $(e^{i\frac{7\pi}{9}})^{-6} (\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})^{-5}$... ??? ... $e^{i\frac{\pi}{2}}$

4.



Braille letter

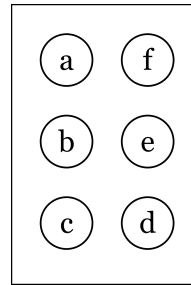
Complex Numbers 1, group $Mu \mu$ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{174} - i^{517} + 5i^{-548} + 3$??? -1
 (b) $2x^2 - x + 2 = 0$??? $\frac{1}{4} \pm \frac{\sqrt{31}}{4}i$
 (c) $(1 - 3i)(7x + yi) = 1 - 4i$??? $3/35$
 (d) $1(-4 + 3i) - (1 + 7i)\overline{(-5 + 4i)}$??? -55 + 42i
 (e) $\left(\frac{-3+8i}{-1-i}\right)^2$??? -24
 (f) $\left(e^{i\frac{3\pi}{5}}\right)^{-2} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-3}$??? $e^{i\frac{13\pi}{10}}$

1.



Braille letter

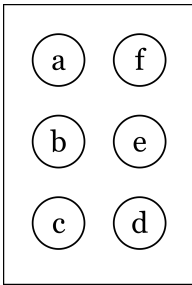
Complex Numbers 1, group $Mu \mu$ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{747} - 8i^{873} - 3i^{-666} + 3$??? -9
 (b) $8x^2 - 3x + 9 = 0$??? $\frac{3}{16} \pm \frac{\sqrt{567}}{16}i$
 (c) $(1 - 2i)(-x + 5yi) = 3 - 2i$.. ??? .. $-31/25$
 (d) $2(-1 - 8i) - (4 + 5i)\overline{(-1 + i)}$??? -3 - 15i
 (e) $\left(\frac{-1+2i}{2-3i}\right)^2$??? $63/169$
 (f) $\left(e^{i\frac{\pi}{6}}\right)^{-2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)^7$... ??? ... $e^{i\frac{5\pi}{12}}$

2.



Braille letter

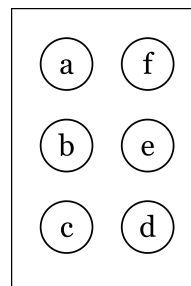
Complex Numbers 1, group $Mu \mu$ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{97} + i^{211} - i^{-967} + 3$??? 1
 (b) $2x^2 - 5x + 6 = 0$??? $\frac{5}{4} \pm \frac{\sqrt{23}}{4}i$
 (c) $(-2 - 2i)(3x + 5yi) = 1 + 2i$... ??? ... $-3/10$
 (d) $3(-9 + i) - (-5 + 4i)\overline{(-2 - 2i)}$??? -9 + 21i
 (e) $\left(\frac{-7+9i}{1-i}\right)^2$??? 63
 (f) $\left(e^{i\pi}\right)^{-1} \left(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7}\right)^{-7}$??? 1

3.



Braille letter

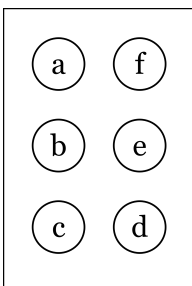
Complex Numbers 1, group $Mu \mu$ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{233} - i^{895} + 6i^{-340} - 7$??? 2
 (b) $2x^2 - 4x + 5 = 0$??? $1 \pm 2i$
 (c) $(4 - 4i)(x + yi) = 6 + 5i$??? $11/4$
 (d) $-4(-5 - 3i) - (-1 - 4i)\overline{(4 - 5i)}$??? -16 + 33i
 (e) $\left(\frac{-1-i}{-2+2i}\right)^2$??? $-1/4$
 (f) $\left(e^{i\frac{4\pi}{3}}\right)^{-7} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{-3}$??? $e^{i\frac{13\pi}{18}}$

4.



Braille letter

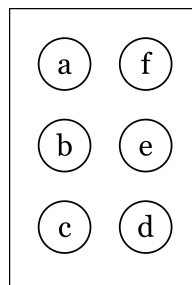
Complex Numbers 1, group $Nu \nu$ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{872} + 4i^{364} - 8i^{-159} - 3$??? -3
 (b) $6x^2 + 2x + 1 = 0$??? $\frac{-1}{6} \pm \frac{\sqrt{20}}{12}i$
 (c) $(-2 - i)(-4x + 3yi) = 8 - 2i$.. ??? .. $3/2$
 (d) $1(9 + 7i) - (3 - 3i)\overline{(-6 + 4i)}$. ??? . $3 + i$
 (e) $\left(\frac{5+2i}{2+i}\right)^2$??? $-29/5$
 (f) $(-8e^{i\frac{7\pi}{6}})^2 (\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4})^{-4}$??? $64e^{i\frac{4\pi}{3}}$

1.



Braille letter

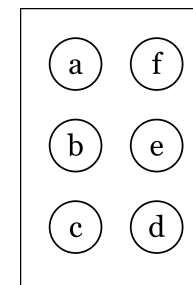
Complex Numbers 1, group $Nu \nu$ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{755} - 3i^{297} + i^{-305} + 3$??? -5
 (b) $7x^2 + 2x + 4 = 0$... ??? ... $\frac{-1}{7} \pm \frac{\sqrt{108}}{14}i$
 (c) $(5 + 3i)(x + 2yi) = -2 + 4i$. ??? . $15/34$
 (d) $3(6 + 5i) - (3 - 4i)\overline{(7 + 5i)}$??? $17 + 28i$
 (e) $\left(\frac{5+4i}{1-i}\right)^2$??? $-41/2$
 (f) $(e^{i\frac{\pi}{5}})^{-5} (\cos \pi + i \sin \pi)^4$... ??? ... $e^{i\frac{36\pi}{35}}$

2.



Braille letter

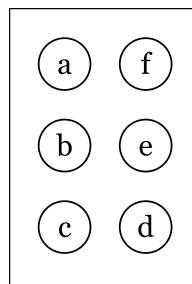
Complex Numbers 1, group $Nu \nu$ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{811} + 2i^{715} + 6i^{-641} - 7$??? -9
 (b) $8x^2 + x + 1 = 0$??? $\frac{-1}{16} \pm \frac{\sqrt{63}}{16}i$
 (c) $(4 + i)(-x - 3yi) = -7 - 3i$??? $98/51$
 (d) $-8(3 - 2i) - (-3 + 3i)\overline{(5 + 9i)}$??? $-66 - 26i$
 (e) $\left(\frac{-1+5i}{-2+2i}\right)^2$??? $5/4$
 (f) $(-5e^{i\frac{\pi}{2}})^2 (\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6})^{-1}$.. ??? .. $25e^{i\frac{\pi}{6}}$

3.



Braille letter

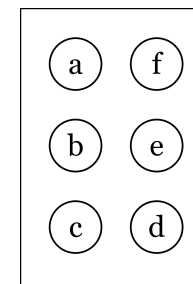
Complex Numbers 1, group $Nu \nu$ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{77} + 2i^{842} + i^{-740} - 2$??? 1
 (b) $4x^2 + 2x + 4 = 0$??? $\frac{-1}{4} \pm \frac{\sqrt{60}}{8}i$
 (c) $(-2 + 4i)(x + 2yi) = -3 - i$... ??? ... $9/20$
 (d) $-5(-1 - 5i) - (-4 - i)\overline{(6 + 4i)}$??? $33 + 15i$
 (e) $\left(\frac{-5-4i}{-2-i}\right)^2$??? $187/25$
 (f) $(e^{i\frac{2\pi}{5}})^{-1} (\cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9})^{-7}$... ??? ... $e^{i\frac{52\pi}{45}}$

4.



Braille letter

Complex Numbers 1, group $Xi \xi$ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{112} + 3i^{75} - i^{-518} + 1$??? -3

(b) $8x^2 - 6x + 2 = 0$??? $\frac{3}{8} \pm \frac{\sqrt{28}}{16}i$

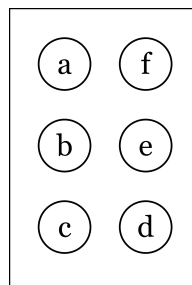
(c) $(2 - 3i)(5x - 2yi) = 1 + i$... ??? ... $-27/130$

(d) $-4(3 - 4i) - (8 + 6i)\overline{(4 - 2i)}$??? $-32 - 24i$

(e) $\left(\frac{-2+4i}{-1-i}\right)^2$??? -8

(f) $\left(-e^{i\frac{\pi}{2}}\right)^5 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{-2}$. ??? . $-e^{i\frac{11\pi}{6}}$

1.



Braille letter

Complex Numbers 1, group $Xi \xi$ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{88} - 2i^{168} + i^{-38} - 6$??? -6

(b) $5x^2 - 5x + 8 = 0$??? $\frac{1}{2} \pm \frac{\sqrt{135}}{10}i$

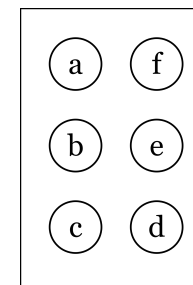
(c) $(4 + 5i)(-x - 3yi) = 1 - 5i$... ??? ... $16/41$

(d) $-5(-1 + 3i) - (-7 + i)\overline{(1 + i)}$??? $-3 - 23i$

(e) $\left(\frac{5-3i}{-1+2i}\right)^2$??? 72

(f) $\left(e^{i\frac{4\pi}{3}}\right)^{-1} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^2$??? $e^{i\pi}$

2.



Braille letter

Complex Numbers 1, group $Xi \xi$ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{28} - 6i^{291} + 6i^{-952} + 5$??? 1

(b) $5x^2 - 7x + 6 = 0$... ??? ... $\frac{7}{10} \pm \frac{\sqrt{71}}{10}i$

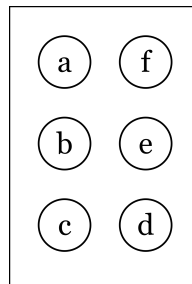
(c) $(-2 - i)(x + yi) = 2 + 5i$. ??? . $-17/5$

(d) $6(6 - 3i) - (2 - i)\overline{(4 - i)}$??? $27 - 12i$

(e) $\left(\frac{-4+3i}{-1-2i}\right)^2$??? -13

(f) $\left(e^{i\frac{\pi}{3}}\right)^{-7} \left(\cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7}\right)^2$??? $e^{i\frac{29\pi}{21}}$

3.



Braille letter

Complex Numbers 1, group $Xi \xi$ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{611} + 2i^{349} + 3i^{-134} + 1$??? 1

(b) $2x^2 + 2x + 1 = 0$??? $-\frac{1}{2} \pm \frac{1}{2}i$

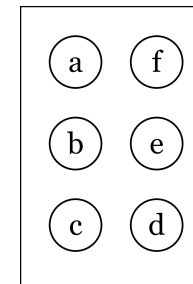
(c) $(2 + 5i)(-2x - yi) = 2 - 4i$. ??? . $6/29$

(d) $8(5 + i) - (4 + 9i)\overline{(-4 + i)}$??? $15 + 48i$

(e) $\left(\frac{-6+2i}{1+i}\right)^2$??? -12

(f) $\left(e^{i\frac{4\pi}{5}}\right)^{-4} \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}\right)^4$. ??? . $e^{i\frac{2\pi}{5}}$

4.



Braille letter

Complex Numbers 1, group *Omicron o* -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{829} + 8i^{581} + 3i^{-62} + 2$??? 9

(b) $8x^2 - 4x + 1 = 0$??? $\frac{1}{4} \pm \frac{\sqrt{48}}{16}i$

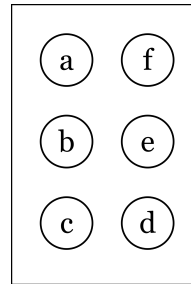
(c) $(3 - 2i)(-2x + 4yi) = 1 + 8i$... ??? ... $1/13$

(d) $-5(3 + 2i) - (7 - 4i)\overline{(-2 - 3i)}$??? $-13 + 3i$

(e) $\left(\frac{-4-5i}{2-i}\right)^2$??? $-41/5$

(f) $\left(e^{i\frac{4\pi}{3}}\right)^{-2} \left(\cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}\right)^{-3}$... ??? ... $e^{i\frac{40\pi}{21}}$

1.



Braille letter

Complex Numbers 1, group *Omicron o* -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{56} + i^{449} + 3i^{-513} + 7$??? -2

(b) $7x^2 + x + 7 = 0$??? $\frac{-1}{14} \pm \frac{\sqrt{195}}{14}i$

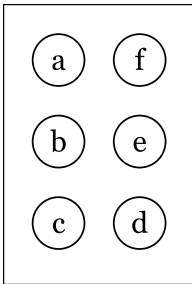
(c) $(-1 + i)(3x + 5yi) = -6 - 5i$. ??? . $19/15$

(d) $3(2 + 3i) - (-3 - 5i)\overline{(6 - i)}$??? $14 + 42i$

(e) $\left(\frac{-1-2i}{-3+i}\right)^2$??? $-1/2$

(f) $\left(-3e^{i\frac{7\pi}{5}}\right)^3 (\cos \pi + i \sin \pi)^3$??? $-27e^{i\frac{5\pi}{4}}$

2.



Braille letter

Complex Numbers 1, group *Omicron o* -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{645} + 7i^{528} + 5i^{-168} - 9$??? 13

(b) $7x^2 + x + 4 = 0$??? $\frac{-1}{14} \pm \frac{\sqrt{111}}{14}i$

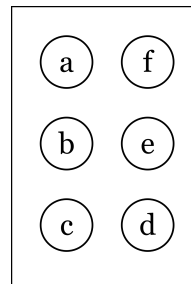
(c) $(3 + i)(5x + yi) = -5 + 2i$... ??? ... $-4/25$

(d) $-7(2 - 2i) - (-1 - 3i)\overline{(8 + 3i)}$??? $3 + 41i$

(e) $\left(\frac{7+5i}{1-i}\right)^2$??? 37

(f) $\left(e^{i\pi}\right)^{-9} (\cos \pi + i \sin \pi)^4$??? $e^{i\pi}$

3.



Braille letter

Complex Numbers 1, group *Omicron o* -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{376} + 3i^{147} - 3i^{-341} + 3$??? 0

(b) $7x^2 + 5x + 4 = 0$??? $\frac{-5}{14} \pm \frac{\sqrt{87}}{14}i$

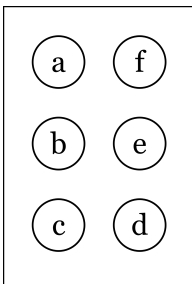
(c) $(-3 - 5i)(-x - 3yi) = -2 - 6i$.. ??? .. $-58/51$

(d) $1(-5 + 3i) - (-1 + 2i)\overline{(-1 + 5i)}$??? $-14 + 0i$

(e) $\left(\frac{1+i}{2-2i}\right)^2$??? $1/4$

(f) $\left(e^{i2\pi}\right)^{-6} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^5$??? $e^{i\frac{\pi}{2}}$

4.



Braille letter

Complex Numbers 1, group $Pi \pi$ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{381} + 7i^{963} + i^{-281} - 9$??? -7

(b) $3x^2 + 2x + 6 = 0$??? $\frac{-1}{3} \pm \frac{\sqrt{68}}{6}i$

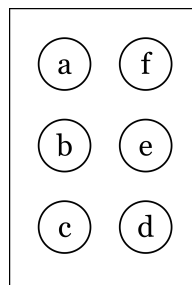
(c) $(-3 + 2i)(-x - 6yi) = -1 - 6i$... ??? ... $17/39$

(d) $-2(-2 + 6i) - (-5 - 3i)\overline{(1 - 6i)}$??? $-19 + 21i$

(e) $\left(\frac{-5+4i}{-2+i}\right)^2$??? $41/5$

(f) $(4e^{i2\pi})^3 (\cos \pi + i \sin \pi)^6$??? 64

1.



Braille letter

Complex Numbers 1, group $Pi \pi$ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{404} + i^{153} + 4i^{-764} - 8$??? 1

(b) $3x^2 - x + 8 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{95}}{6}i$

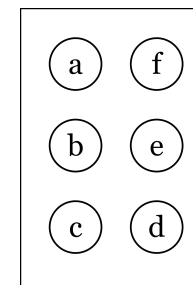
(c) $(3 - 4i)(-3x - yi) = -9 + 2i$.. ??? .. $5/3$

(d) $5(2 + i) - (-5 + 9i)\overline{(2 + 7i)}$??? $-43 + 22i$

(e) $\left(\frac{-2-2i}{-1-3i}\right)^2$??? $4/5$

(f) $(e^{i\frac{6\pi}{7}})^{-3} (\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})^{-4}$.. ??? .. $e^{i\frac{81\pi}{56}}$

2.



Braille letter

Complex Numbers 1, group $Pi \pi$ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{961} + 4i^{509} - 5i^{-709} - 1$??? 10

(b) $2x^2 - 6x + 7 = 0$??? $\frac{3}{2} \pm \frac{\sqrt{20}}{2}i$

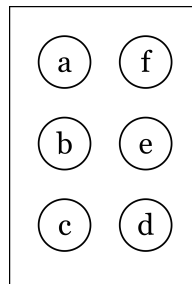
(c) $(5 - 4i)(-3x + yi) = 3 + 4i$.. ??? .. $65/123$

(d) $6(5 + 2i) - (-6 + 7i)\overline{(6 + 2i)}$??? $38 - 42i$

(e) $\left(\frac{5-4i}{2+2i}\right)^2$??? $-41/8$

(f) $(8e^{i\frac{\pi}{2}})^3 (\cos 2\pi + i \sin 2\pi)^5$. ??? . $512e^{i\frac{13\pi}{8}}$

3.



Braille letter

Complex Numbers 1, group $Pi \pi$ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{389} - 7i^{181} + 6i^{-162} + 6$??? -6

(b) $3x^2 - x + 7 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{83}}{3}i$

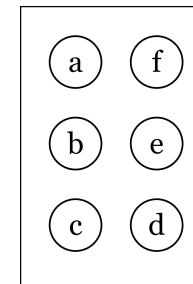
(c) $(2 - 3i)(-x + 6yi) = 8 + i$... ??? ... $-2/3$

(d) $-1(3 - 4i) - (9 + i)\overline{(-3 - 3i)}$??? $27 + 34i$

(e) $\left(\frac{2-2i}{-1+i}\right)^2$??? 4

(f) $(e^{i\frac{\pi}{3}})^{-4} (\cos 2\pi + i \sin 2\pi)^{-6}$.. ??? .. $e^{i\frac{2\pi}{3}}$

4.



Braille letter

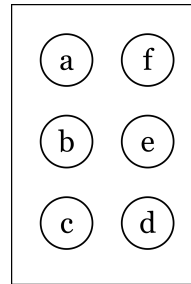
Complex Numbers 1, group *Rho* ρ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{556} + 8i^{31} - 3i^{-343} - 2$??? -11
 (b) $9x^2 - 6x + 5 = 0$??? $\frac{1}{3} \pm \frac{4}{3}i$
 (c) $(4 - 4i)(-x - 2yi) = -2 - 4i$.. ??? .. $1/8$
 (d) $8(2 + 5i) - (1 + 3i)\overline{(-2 - 6i)}$??? $36 + 52i$
 (e) $\left(\frac{8+6i}{-1+i}\right)^2$??? -48
 (f) $\left(e^{i\frac{3\pi}{4}}\right)^{-2} (\cos 2\pi + i \sin 2\pi)^{-6}$.. ??? .. $e^{i\frac{\pi}{2}}$

1.



Braille letter

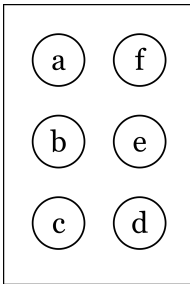
Complex Numbers 1, group *Rho* ρ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{556} + 2i^{780} + i^{-785} + 4$??? 4
 (b) $x^2 + x + 8 = 0$??? $\frac{-1}{2} \pm \frac{\sqrt{31}}{2}i$
 (c) $(-4 + i)(5x + yi) = -1 - 3i$??? $72/85$
 (d) $2(-7 - 6i) - (-2 + 4i)\overline{(-6 + 5i)}$??? $-46 + 22i$
 (e) $\left(\frac{-2-i}{-1-2i}\right)^2$??? -1
 (f) $(3e^{i\pi})^6 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{-1}$... ??? ... $729e^{i\frac{4\pi}{3}}$

2.



Braille letter

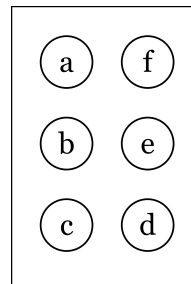
Complex Numbers 1, group *Rho* ρ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{845} - 9i^{51} - 2i^{-205} + 6$??? 12
 (b) $5x^2 - 8x + 7 = 0$??? $\frac{4}{5} \pm \frac{\sqrt{76}}{5}i$
 (c) $(3 + 5i)(-x + yi) = 2 + 5i$??? $-13/17$
 (d) $6(-2 - 8i) - (-9 - 2i)\overline{(-9 + 7i)}$??? $83 - 129i$
 (e) $\left(\frac{-3-2i}{1+3i}\right)^2$??? $8/25$
 (f) $(-4e^{i2\pi})^4 \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}\right)^7$... ??? ... $256e^{i\frac{\pi}{5}}$

3.



Braille letter

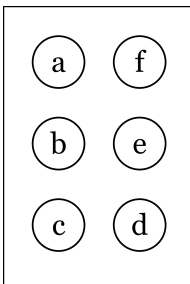
Complex Numbers 1, group *Rho* ρ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{99} + 2i^{391} + 8i^{-884} + 4$??? -3
 (b) $5x^2 + 2x + 1 = 0$??? $\frac{-1}{5} \pm \frac{4}{5}i$
 (c) $(2 - i)(-6x - 3yi) = -7 - i$??? $1/10$
 (d) $-7(-2 + 3i) - (1 - 7i)\overline{(-4 - 7i)}$??? $-39 - 56i$
 (e) $\left(\frac{1+2i}{2-i}\right)^2$??? -1
 (f) $(-e^{i\frac{3\pi}{5}})^8 \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)^{-2}$??? $e^{i\frac{13\pi}{10}}$

4.



Braille letter

Complex Numbers 1, group *Sigma* σ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{315} - 6i^{961} + 2i^{-124} + 1$??? -7

(b) $5x^2 + 7x + 3 = 0$??? $\frac{-7}{10} \pm \frac{\sqrt{11}}{10}i$

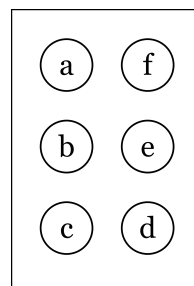
(c) $(5 + 2i)(3x + yi) = 7 - 3i$??? $-46/87$

(d) $-3(-6 - 3i) - (4 + 2i)\overline{(-6 - 2i)}$??? $-2 + 13i$

(e) $\left(\frac{-4+5i}{1-i}\right)^2$??? $-41/2$

(f) $\left(e^{i\frac{3\pi}{2}}\right)^{-3} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^{-5}$??? $e^{i\frac{2\pi}{3}}$

1.



Braille letter

Complex Numbers 1, group *Sigma* σ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{892} + 4i^{534} + i^{-315} + 3$??? 1

(b) $6x^2 - 2x + 6 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{284}}{12}i$

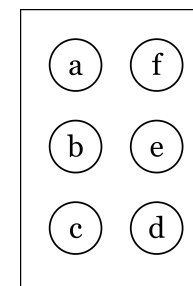
(c) $(3 - 3i)(-x + 2yi) = 1 - 2i$??? $-1/4$

(d) $2(2 - 2i) - (-2 - 8i)\overline{(-2 - 5i)}$??? $-80 - 10i$

(e) $\left(\frac{-9+2i}{1-i}\right)^2$??? $85/2$

(f) $\left(e^{i\frac{2\pi}{5}}\right)^{-5} \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}\right)^{-2}$... ??? ... $e^{i\frac{4\pi}{5}}$

2.



Braille letter

Complex Numbers 1, group *Sigma* σ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{184} - 2i^{77} + 3i^{-567} - 6$??? 1

(b) $5x^2 - x + 6 = 0$??? $\frac{1}{10} \pm \frac{\sqrt{119}}{10}i$

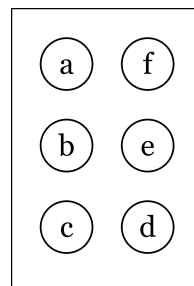
(c) $(-1 - 2i)(-2x - 3yi) = 3 + 4i$.. ??? .. $29/30$

(d) $-2(1 + 8i) - (-6 - 5i)\overline{(6 + 3i)}$??? $-23 - 4i$

(e) $\left(\frac{2-i}{3+2i}\right)^2$??? $-33/169$

(f) $\left(-2e^{i2\pi}\right)^2 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)^3$... ??? ... $4e^{i\pi}$

3.



Braille letter

Complex Numbers 1, group *Sigma* σ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{308} - 2i^{394} - 7i^{-38} + 2$??? 0

(b) $5x^2 + x + 1 = 0$??? $\frac{-1}{10} \pm \frac{\sqrt{39}}{10}i$

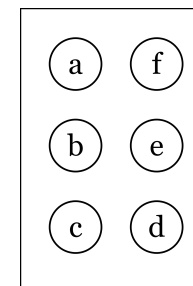
(c) $(-3 + 2i)(-x - 3yi) = -4 + 2i$. ??? . $-50/39$

(d) $-2(4 - 4i) - (3 + 2i)\overline{(-1 - 5i)}$??? $-1 - 5i$

(e) $\left(\frac{-3+2i}{-2-3i}\right)^2$??? $-169/25$

(f) $\left(-2e^{i\frac{5\pi}{7}}\right)^5 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)^{-8}$??? $-32e^{i\frac{11\pi}{7}}$

4.



Braille letter

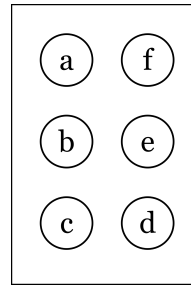
Complex Numbers 1, group *Tau* τ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{718} - 2i^{756} + i^{-582} - 3$??? 0
 (b) $3x^2 - 5x + 6 = 0$??? $\frac{5}{6} \pm \frac{\sqrt{47}}{6}i$
 (c) $(-4 - 5i)(-4x + yi) = -4 + 6i$??? $-81/82$
 (d) $3(-2 - 7i) - (-5 + 6i)\overline{(6 + i)}$??? $18 - 62i$
 (e) $\left(\frac{3-5i}{-2+2i}\right)^2$??? $15/4$
 (f) $(-2e^{i\frac{\pi}{5}})^3 (\cos \pi + i \sin \pi)^{-7}$. ??? . $-8e^{i\frac{9\pi}{5}}$

1.



Braille letter

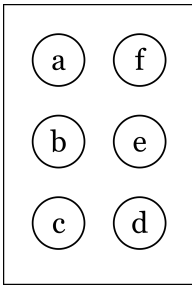
Complex Numbers 1, group *Tau* τ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{695} - 3i^{704} + 2i^{-836} - 2$... ??? ... -1
 (b) $3x^2 - x + 4 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{95}}{6}i$
 (c) $(-5 - 2i)(x - yi) = -2 - 4i$??? $-6/29$
 (d) $3(4 - 4i) - (3 + i)\overline{(-3 + i)}$??? $19 - 6i$
 (e) $\left(\frac{-9+6i}{-1-i}\right)^2$??? -24
 (f) $(e^{i\frac{2\pi}{3}})^{-2} (\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8})^{-4}$??? $e^{i\frac{\pi}{6}}$

2.



Braille letter

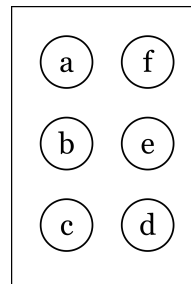
Complex Numbers 1, group *Tau* τ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{866} - 7i^{569} + 8i^{-721} + 9$??? -15
 (b) $3x^2 - x + 5 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{59}}{6}i$
 (c) $(-3 - 3i)(x + 5yi) = -2 - 4i$.. ??? .. $16/15$
 (d) $7(3 - i) - (-6 + 3i)\overline{(-3 - 8i)}$??? $51 + 50i$
 (e) $\left(\frac{-3-4i}{-2-i}\right)^2$??? 5
 (f) $(e^{i\frac{3\pi}{4}})^{-5} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})^{-4}$.. ??? .. $e^{i\frac{21\pi}{16}}$

3.



Braille letter

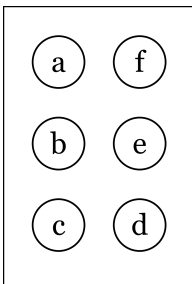
Complex Numbers 1, group *Tau* τ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{362} + i^{648} - 4i^{-833} - 1$??? 4
 (b) $6x^2 - 2x + 9 = 0$??? $\frac{1}{6} \pm \frac{\sqrt{212}}{12}i$
 (c) $(3 + i)(-2x - 3yi) = -3 - 8i$.. ??? .. $31/20$
 (d) $-4(-1 - i) - (8 + 8i)\overline{(2 - 6i)}$??? $84 - 60i$
 (e) $\left(\frac{6-6i}{1-i}\right)^2$??? 16
 (f) $(-3e^{i\frac{3\pi}{2}})^2 (\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8})^4$. ??? . $9e^{i\frac{3\pi}{2}}$

4.



Braille letter

Complex Numbers 1, group *Upsilon v* -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{555} + i^{66} + 7i^{-599} - 2$??? 6

(b) $2x^2 - x + 1 = 0$??? $\frac{1}{4} \pm \frac{\sqrt{7}}{4}i$

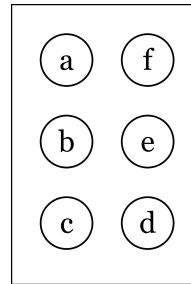
(c) $(-1 - 3i)(7x + yi) = 2 + 5i$??? $2/7$

(d) $-3(7 + 4i) - (5 + i)\overline{(-3 - 2i)}$??? $-34 - 19i$

(e) $\left(\frac{-3-5i}{1+i}\right)^2$??? $20/3$

(f) $(-3e^{i\frac{4\pi}{5}})^5 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-2}$. ??? . $-243e^{i\pi}$

1.



Braille letter

Complex Numbers 1, group *Upsilon v* -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{492} - 7i^{390} + 8i^{-813} + 6$??? -8

(b) $6x^2 + 4x + 6 = 0$... ??? ... $\frac{-1}{3} \pm \frac{\sqrt{272}}{12}i$

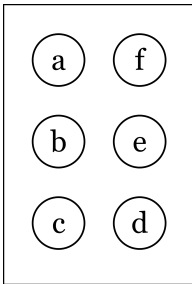
(c) $(1 - 7i)(x + 2yi) = 1 + i$... ??? ... $-1/25$

(d) $3(5 + 3i) - (-7 - 5i)\overline{(-1 - i)}$??? $3 - 3i$

(e) $\left(\frac{1-5i}{2-i}\right)^2$??? $-32/25$

(f) $(e^{i\frac{5\pi}{3}})^{-2} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-4}$.. ??? .. $e^{i\frac{3\pi}{4}}$

2.



Braille letter

Complex Numbers 1, group *Upsilon v* -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{640} + 2i^{188} - 4i^{-516} + 8$??? 0

(b) $4x^2 + 3x + 5 = 0$??? $\frac{-3}{8} \pm \frac{\sqrt{151}}{8}i$

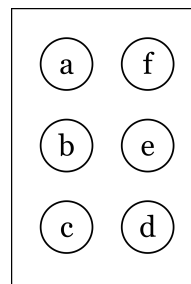
(c) $(3 + i)(-3x - yi) = 1 + 6i$... ??? ... -2

(d) $3(1 + 4i) - (-5 + 4i)\overline{(6 - 7i)}$??? $89 + 23i$

(e) $\left(\frac{-2-6i}{1-i}\right)^2$??? -12

(f) $(e^{i\frac{3\pi}{2}})^{-3} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^{-4}$.. ??? .. $e^{i\frac{13\pi}{8}}$

3.



Braille letter

Complex Numbers 1, group *Upsilon v* -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{206} + 2i^{770} + 2i^{-97} + 5$??? -2

(b) $9x^2 + 3x + 3 = 0$??? $\frac{-1}{6} \pm \frac{\sqrt{99}}{9}i$

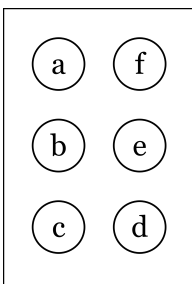
(c) $(-1 + 2i)(-2x + 2yi) = -6 + 2i$. ??? . $4/5$

(d) $1(-5 + 3i) - (-1 - 6i)\overline{(2 + 2i)}$??? $9 + 17i$

(e) $\left(\frac{-5-9i}{1-i}\right)^2$??? -45

(f) $(-3e^{i\pi})^4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{-9}$.. ??? .. $81e^{i\pi}$

4.



Braille letter

Complex Numbers 1, group *Phi* ϕ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{310} - 4i^{642} + 7i^{-621} - 2$??? 4

(b) $2x^2 + 3x + 5 = 0$??? $-\frac{3}{4} \pm \frac{\sqrt{31}}{4}i$

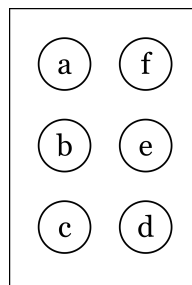
(c) $(4 - i)(-x - yi) = -2 - 9i$??? $37/17$

(d) $-2(-5 - 9i) - (-1 + 7i)\overline{(-4 + 6i)}$??? $-28 + 40i$

(e) $\left(\frac{3-6i}{-1+i}\right)^2$??? $45/2$

(f) $\left(e^{i\frac{\pi}{3}}\right)^{-3} (\cos \pi + i \sin \pi)^{-5}$??? 1

1.



Braille letter

Complex Numbers 1, group *Phi* ϕ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{97} + 2i^{124} + 3i^{-191} + 9$??? 6

(b) $3x^2 + 6x + 7 = 0$??? $-1 \pm \frac{\sqrt{48}}{6}i$

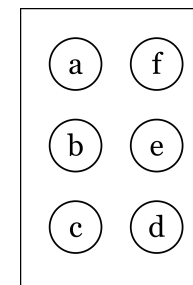
(c) $(-4 + 2i)(x + 3yi) = 2 + 4i$??? $-1/3$

(d) $-1(5 - 2i) - (-6 + 8i)\overline{(-6 - 4i)}$??? $63 + 74i$

(e) $\left(\frac{4+4i}{1-i}\right)^2$??? -16

(f) $\left(e^{i\frac{3\pi}{2}}\right)^{-1} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)^{-3}$??? $e^{i\frac{7\pi}{4}}$

2.



Braille letter

Complex Numbers 1, group *Phi* ϕ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{283} + 2i^{976} - 4i^{-322} + 6$??? -1

(b) $2x^2 - 3x + 2 = 0$??? $\frac{3}{4} \pm \frac{\sqrt{7}}{2}i$

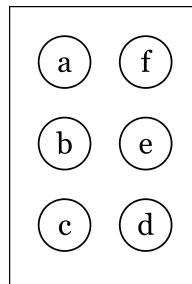
(c) $(4 - 7i)(x - yi) = -3 - 2i$... ??? ... $-11/65$

(d) $-2(7 - i) - (-2 - 7i)\overline{(-1 - 2i)}$??? $-26 - i$

(e) $\left(\frac{-3-2i}{2+i}\right)^2$??? 7

(f) $\left(e^{i\frac{\pi}{2}}\right)^{-1} \left(\cos \frac{5\pi}{9} + i \sin \frac{5\pi}{9}\right)^{-8}$... ??? ... $e^{i\frac{35\pi}{18}}$

3.



Braille letter

Complex Numbers 1, group *Phi* ϕ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{769} - 6i^{775} + 6i^{-264} - 7$??? 7

(b) $x^2 - 3x + 5 = 0$??? $\frac{3}{2} \pm \frac{\sqrt{11}}{2}i$

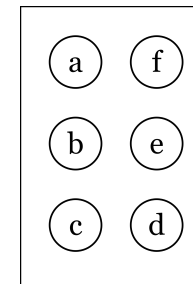
(c) $(2 + 3i)(3x - 3yi) = 6 + 3i$... ??? ... $11/13$

(d) $-1(-2 - i) - (1 + i)\overline{(-7 + i)}$. ??? . $7 + 9i$

(e) $\left(\frac{-2-i}{1+3i}\right)^2$??? 0

(f) $\left(-8e^{i\frac{4\pi}{3}}\right)^3 \left(\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}\right)^2$??? $-512e^{i\frac{2\pi}{5}}$

4.



Braille letter

Complex Numbers 1, group *Chi* χ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{213} + i^{38} + 2i^{-806} + 1$??? 1

(b) $8x^2 - 3x + 5 = 0$... ??? ... $\frac{3}{16} \pm \frac{\sqrt{311}}{16}i$

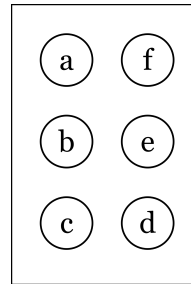
(c) $(-2 + i)(4x - 5yi) = 3 - 7i$. ??? . $3/100$

(d) $1(6 + 3i) - (-7 - 5i)\overline{(2 - i)}$??? $15 + 6i$

(e) $\left(\frac{4+3i}{1+i}\right)^2$??? $16/3$

(f) $\left(e^{i\frac{2\pi}{3}}\right)^{-3} \left(\cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7}\right)^5$. ??? . $e^{i\frac{2\pi}{7}}$

1.



Braille letter

Complex Numbers 1, group *Chi* χ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{514} + 3i^{457} - 7i^{-561} - 2$??? 10

(b) $4x^2 - 3x + 2 = 0$??? $\frac{3}{8} \pm \frac{\sqrt{23}}{8}i$

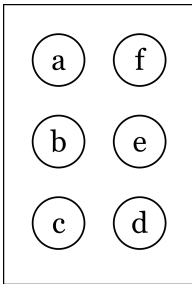
(c) $(-1 + 2i)(-4x - 6yi) = 7 - 3i$.. ??? .. $61/60$

(d) $3(-1 + 5i) - (-5 + 2i)\overline{(-7 + 9i)}$??? $14 - 16i$

(e) $\left(\frac{-5+5i}{-1-i}\right)^2$??? 25

(f) $\left(e^{i\frac{\pi}{2}}\right)^{-3} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^2$??? $e^{i\frac{11\pi}{6}}$

2.



Braille letter

Complex Numbers 1, group *Chi* χ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{505} - i^{691} - i^{-830} + 4$??? 2

(b) $5x^2 - x + 8 = 0$??? $\frac{1}{10} \pm \frac{\sqrt{159}}{5}i$

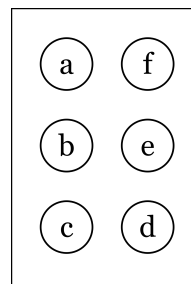
(c) $(1 - 9i)(x - yi) = 1 - 2i$??? $6/41$

(d) $2(-3 + 8i) - (-1 - 9i)\overline{(2 - i)}$??? $-13 + 35i$

(e) $\left(\frac{-3-4i}{1+i}\right)^2$??? $16/3$

(f) $\left(-e^{i\frac{5\pi}{6}}\right)^3 \left(\cos \frac{5\pi}{7} + i \sin \frac{5\pi}{7}\right)^6$.. ??? .. $-e^{i\frac{3\pi}{14}}$

3.



Braille letter

Complex Numbers 1, group *Chi* χ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{598} - 2i^{814} + i^{-394} + 5$??? 0

(b) $x^2 + 3x + 7 = 0$??? $-\frac{3}{2} \pm \frac{\sqrt{19}}{2}i$

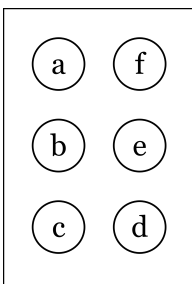
(c) $(-1 - 3i)(-x - 5yi) = -1 + 2i$... ??? ... $-3/5$

(d) $-7(4 + 2i) - (-3 - 2i)\overline{(-3 - 2i)}$??? $-23 - 14i$

(e) $\left(\frac{3+4i}{-1-2i}\right)^2$??? 13

(f) $\left(e^{i\frac{\pi}{2}}\right)^{-3} (\cos \pi + i \sin \pi)^3$??? $e^{i2\pi}$

4.



Braille letter

Complex Numbers 1, group *Psi* ψ -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{226} - i^{652} + 3i^{-991} - 6$??? 3

(b) $8x^2 - 5x + 9 = 0$??? $\frac{5}{16} \pm \frac{\sqrt{263}}{16}i$

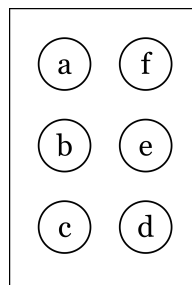
(c) $(-3 - 3i)(-x - 3yi) = -4 - 7i$.. ??? .. -2

(d) $-6(-4 + 8i) - (6 + 3i)\overline{(-1 + i)}$??? 27 - 39i

(e) $\left(\frac{1+4i}{-2+i}\right)^2$??? -77/25

(f) $\left(e^{i\frac{8\pi}{5}}\right)^{-1} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^8$??? $e^{i\frac{13\pi}{30}}$

1.



Braille letter

Complex Numbers 1, group *Psi* ψ -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{778} + 5i^{446} + 4i^{-189} - 3$??? 10

(b) $3x^2 - 5x + 9 = 0$??? $\frac{5}{6} \pm \frac{\sqrt{83}}{6}i$

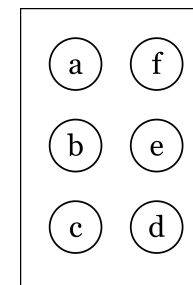
(c) $(4 - 4i)(4x - yi) = -1 + 4i$... ??? ... 15/32

(d) $8(1 + 6i) - (3 - 7i)\overline{(-7 - 9i)}$??? -76 - 28i

(e) $\left(\frac{-2+3i}{3-2i}\right)^2$??? 1

(f) $\left(e^{i\frac{\pi}{3}}\right)^{-1} \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)^2$??? $e^{i\frac{\pi}{3}}$

2.



Braille letter

Complex Numbers 1, group *Psi* ψ -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{783} + 2i^{992} - 4i^{-848} + 3$??? -1

(b) $7x^2 + 3x + 2 = 0$??? $\frac{-3}{14} \pm \frac{\sqrt{103}}{14}i$

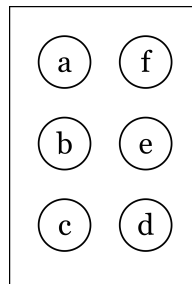
(c) $(-2 + 2i)(x - 4yi) = 4 + 2i$.. ??? .. -1/8

(d) $1(1 - 9i) - (2 + 3i)\overline{(-1 - 5i)}$??? 33 - 16i

(e) $\left(\frac{-1-5i}{-2+2i}\right)^2$??? -5/4

(f) $\left(e^{i\frac{\pi}{2}}\right)^{-1} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{-6}$... ??? ... $e^{i\frac{3\pi}{2}}$

3.



Braille letter

Complex Numbers 1, group *Psi* ψ -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

(a) $i^{635} + 5i^{126} - 4i^{-200} + 7$??? -1

(b) $2x^2 - 2x + 2 = 0$??? $\frac{1}{2} \pm \frac{\sqrt{12}}{4}i$

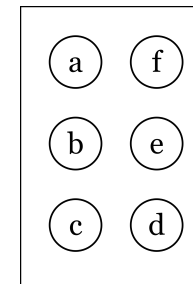
(c) $(1 - 2i)(-7x - 3yi) = -2 - 6i$??? -16/105

(d) $-1(2 + 6i) - (2 + 2i)\overline{(7 + 3i)}$??? -22 - 26i

(e) $\left(\frac{-5-4i}{1-i}\right)^2$??? -20

(f) $\left(e^{i\frac{\pi}{3}}\right)^{-7} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{-5}$... ??? ... $e^{i\frac{\pi}{3}}$

4.



Braille letter

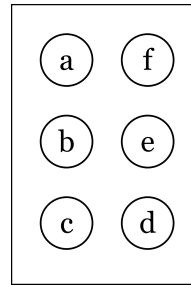
Complex Numbers 1, group *Omega* ω -i

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{87} - 2i^{940} - i^{-951} + 4$??? -2
 (b) $3x^2 + 2x + 5 = 0$??? $-\frac{1}{3} \pm \frac{\sqrt{116}}{6}i$
 (c) $(3 - 2i)(-2x + yi) = 2 + 3i$.. ??? .. $5/13$
 (d) $-4(8 + 9i) - (4 - 3i)\overline{(8 - i)}$??? -3 - 16i
 (e) $\left(\frac{-6+6i}{-1+i}\right)^2$??? 16
 (f) $(-6e^{i2\pi})^2 (\cos 2\pi + i \sin 2\pi)^7$.. ??? .. 36

1.



Braille letter

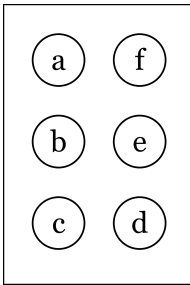
Complex Numbers 1, group *Omega* ω -ii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{310} - i^{889} - 2i^{-373} - 2$??? 1
 (b) $3x^2 - 2x + 8 = 0$??? $\frac{1}{3} \pm \frac{\sqrt{92}}{6}i$
 (c) $(-1 - 4i)(-5x - yi) = 2 + 6i$... ??? ... $96/85$
 (d) $4(-4 - 6i) - (-5 + 4i)\overline{(5 + 4i)}$??? -23 - 64i
 (e) $\left(\frac{-5-7i}{-1-i}\right)^2$??? 35
 (f) $(e^{i\frac{\pi}{4}})^{-3} (\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})^{-4}$??? $e^{i\frac{31\pi}{24}}$

2.



Braille letter

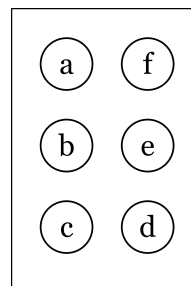
Complex Numbers 1, group *Omega* ω -iii

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{287} - i^{820} + i^{-323} + 7$??? 0
 (b) $2x^2 - x + 5 = 0$??? $\frac{1}{4} \pm \frac{\sqrt{79}}{4}i$
 (c) $(2 + 2i)(-4x + 4yi) = 1 + 8i$. ??? . -9/8
 (d) $4(-3 - 3i) - (1 + 2i)\overline{(-2 - 7i)}$??? 4 - i
 (e) $\left(\frac{4-4i}{-1-i}\right)^2$??? 16
 (f) $(e^{i\frac{8\pi}{9}})^{-2} (\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})^3$.. ??? .. $e^{i\frac{7\pi}{27}}$

3.



Braille letter

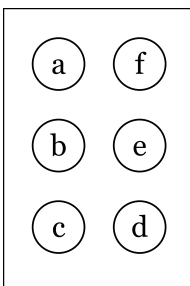
Complex Numbers 1, group *Omega* ω -iv

Name:

- (a) Determine the imaginary part.
 (b) Calculate the roots of the quadratic equation in the field of complex numbers.
 (c) Find the sum of the real numbers x and y that satisfy the equation.
 (d) Write in the algebraic form.
 (e) Determine the real part of the complex number.
 (f) Simplify and write the resulting complex number in the exponential form.

- (a) $i^{432} - 4i^{787} + i^{-962} + 3$??? -2
 (b) $8x^2 - 2x + 1 = 0$??? $\frac{1}{8} \pm \frac{\sqrt{28}}{16}i$
 (c) $(-5 + 6i)(2x - yi) = -6 - i$. ??? . -29/61
 (d) $2(-4 + 4i) - (3 - 3i)\overline{(2 + 3i)}$??? 7 + 23i
 (e) $\left(\frac{-1-2i}{3+i}\right)^2$??? 0
 (f) $(e^{i\frac{3\pi}{2}})^{-7} (\cos 2\pi + i \sin 2\pi)^{-2}$. ??? . $e^{i\frac{3\pi}{2}}$

4.



Braille letter

Complex Numbers 1 (solutions)

α	$i : \mathbf{M}$	(a) $-3\checkmark$	(b) $-2/7 \pm \sqrt{96}/14i\checkmark$	(c) $-23/15\checkmark$	(d) $-19 + 17i\checkmark$	(e) $2/9\checkmark$	(f) $e^{i4\pi/3}\checkmark$
	$ii : \mathbf{E}$	(a) $9\checkmark$	(b) $1/10 \pm \sqrt{119}/10i\checkmark$	(c) $19/39\checkmark$	(d) $-32 - 26i\checkmark$	(e) $-24/169\checkmark$	(f) $e^{i\pi/2}\checkmark$
	$iii : \mathbf{E}$	(a) $-3\checkmark$	(b) $-5/6 \pm \sqrt{59}/6i\checkmark$	(c) $-31/73\checkmark$	(d) $-20 + 47i\checkmark$	(e) $32/169\checkmark$	(f) $1\checkmark$
	$iv : \mathbf{T}$	(a) $0\checkmark$	(b) $1 \pm \sqrt{24}/4i\checkmark$	(c) $-25/39\checkmark$	(d) $21 - 26i\checkmark$	(e) $1/4\checkmark$	(f) $-32e^{i5\pi/4}\checkmark$
	$i : \mathbf{L}$	(a) $-6\checkmark$	(b) $1/6 \pm \sqrt{11}/6i\checkmark$	(c) $1\checkmark$	(d) $-22 - 70i\checkmark$	(e) $-48/25\checkmark$	(f) $243e^{i7\pi/4}\checkmark$
β	$ii : \mathbf{A}$	(a) $4\checkmark$	(b) $3/5 \pm 1/5i\checkmark$	(c) $3/2\checkmark$	(d) $12 + 26i\checkmark$	(e) $28\checkmark$	(f) $e^{i3\pi/4}\checkmark$
	$iii : \mathbf{N}$	(a) $-5\checkmark$	(b) $-2 \pm i\checkmark$	(c) $-11/15\checkmark$	(d) $13 + 35i\checkmark$	(e) $-72/25\checkmark$	(f) $16e^{i2\pi/15}\checkmark$
	$iv : \mathbf{D}$	(a) $-3\checkmark$	(b) $-1/16 \pm \sqrt{191}/16i\checkmark$	(c) $-5/3\checkmark$	(d) $12 - 16i\checkmark$	(e) $-117/25\checkmark$	(f) $36e^{i\pi}\checkmark$
	$i : \mathbf{C}$	(a) $-7\checkmark$	(b) $-3/4 \pm \sqrt{12}/8i\checkmark$	(c) $3/5\checkmark$	(d) $-21 + 30i\checkmark$	(e) $-72/25\checkmark$	(f) $1\checkmark$
	$ii : \mathbf{A}$	(a) $-3\checkmark$	(b) $-1/2 \pm \sqrt{115}/10i\checkmark$	(c) $-37/39\checkmark$	(d) $5 - 9i\checkmark$	(e) $143/25\checkmark$	(f) $e^{i\pi}\checkmark$
γ	$iii : \mathbf{S}$	(a) $-3\checkmark$	(b) $2/7 \pm \sqrt{96}/14i\checkmark$	(c) $-13/40\checkmark$	(d) $-17 + 75i\checkmark$	(e) $64\checkmark$	(f) $e^{i17\pi/21}\checkmark$
	$iv : \mathbf{H}$	(a) $-1\checkmark$	(b) $3/8 \pm \sqrt{23}/8i\checkmark$	(c) $-1/10\checkmark$	(d) $-1 - 24i\checkmark$	(e) $-4\checkmark$	(f) $e^{i\pi/5}\checkmark$
	$i : \mathbf{V}$	(a) $-2\checkmark$	(b) $-1/8 \pm \sqrt{220}/16i\checkmark$	(c) $1/5\checkmark$	(d) $-25 + 11i\checkmark$	(e) $1/9\checkmark$	(f) $1\checkmark$
	$ii : \mathbf{I}$	(a) $0\checkmark$	(b) $-1/2 \pm \sqrt{19}/2i\checkmark$	(c) $-5/34\checkmark$	(d) $7 - 34i\checkmark$	(e) $-17/25\checkmark$	(f) $e^{i5\pi/3}\checkmark$
	$iii : \mathbf{E}$	(a) $2\checkmark$	(b) $3/7 \pm \sqrt{132}/14i\checkmark$	(c) $-89/28\checkmark$	(d) $1 + 11i\checkmark$	(e) $-16\checkmark$	(f) $e^{i3\pi/7}\checkmark$
δ	$iv : \mathbf{W}$	(a) $5\checkmark$	(b) $1/2 \pm \sqrt{28}/4i\checkmark$	(c) $17/20\checkmark$	(d) $57 - 28i\checkmark$	(e) $-1\checkmark$	(f) $e^{i\pi/3}\checkmark$
	$i : \mathbf{H}$	(a) $0\checkmark$	(b) $1/3 \pm \sqrt{68}/6i\checkmark$	(c) $-9/5\checkmark$	(d) $-34 - 26i\checkmark$	(e) $-18\checkmark$	(f) $e^{i9\pi/7}\checkmark$
	$ii : \mathbf{O}$	(a) $5\checkmark$	(b) $3/8 \pm \sqrt{39}/8i\checkmark$	(c) $-1\checkmark$	(d) $18 - 9i\checkmark$	(e) $-6\checkmark$	(f) $1\checkmark$
	$iii : \mathbf{P}$	(a) $9\checkmark$	(b) $-1/4 \pm 3/4i\checkmark$	(c) $4/15\checkmark$	(d) $7 - 15i\checkmark$	(e) $8\checkmark$	(f) $e^{i\pi/3}\checkmark$
	$iv : \mathbf{E}$	(a) $0\checkmark$	(b) $-1/4 \pm \sqrt{124}/8i\checkmark$	(c) $-2/5\checkmark$	(d) $18 + 42i\checkmark$	(e) $-12/25\checkmark$	(f) $e^{i11\pi/20}\checkmark$
ζ	$i : \mathbf{G}$	(a) $-6\checkmark$	(b) $3/4 \pm \sqrt{15}/4i\checkmark$	(c) $85/116\checkmark$	(d) $-37 + 16i\checkmark$	(e) $2/9\checkmark$	(f) $81\checkmark$
	$ii : \mathbf{R}$	(a) $3\checkmark$	(b) $-1/4 \pm \sqrt{23}/4i\checkmark$	(c) $-27/29\checkmark$	(d) $-71 + 31i\checkmark$	(e) $0\checkmark$	(f) $e^{i5\pi/4}\checkmark$
	$iii : \mathbf{A}$	(a) $0\checkmark$	(b) $-7/16 \pm \sqrt{47}/16i\checkmark$	(c) $54/65\checkmark$	(d) $11 - 31i\checkmark$	(e) $-12/25\checkmark$	(f) $243e^{i11\pi/7}\checkmark$
	$iv : \mathbf{Y}$	(a) $1\checkmark$	(b) $1/2 \pm \sqrt{27}/2i\checkmark$	(c) $16/53\checkmark$	(d) $0 - 48i\checkmark$	(e) $72/25\checkmark$	(f) $e^{i11\pi/10}\checkmark$
	$i : \mathbf{S}$	(a) $2\checkmark$	(b) $1/14 \pm \sqrt{55}/14i\checkmark$	(c) $-6/17\checkmark$	(d) $6 + 40i\checkmark$	(e) $-1\checkmark$	(f) $4e^{i3\pi/2}\checkmark$
η	$ii : \mathbf{E}$	(a) $1\checkmark$	(b) $-1/4 \pm \sqrt{31}/4i\checkmark$	(c) $-41/37\checkmark$	(d) $11 - 28i\checkmark$	(e) $-14\checkmark$	(f) $-64e^{i8\pi/7}\checkmark$
	$iii : \mathbf{A}$	(a) $12\checkmark$	(b) $1/2 \pm \sqrt{3}/2i\checkmark$	(c) $33/25\checkmark$	(d) $-1 - 28i\checkmark$	(e) $-77/25\checkmark$	(f) $e^{i48\pi/35}\checkmark$
	$iv : \mathbf{T}$	(a) $0\checkmark$	(b) $3/16 \pm \sqrt{215}/16i\checkmark$	(c) $-25/38\checkmark$	(d) $-10 + 22i\checkmark$	(e) $-3/4\checkmark$	(f) $e^{i3\pi/2}\checkmark$
	$i : \mathbf{T}$	(a) $-9\checkmark$	(b) $-1/8 \pm \sqrt{28}/16i\checkmark$	(c) $-7/4\checkmark$	(d) $34 - 16i\checkmark$	(e) $10\checkmark$	(f) $e^{i\pi/4}\checkmark$
	$ii : \mathbf{I}$	(a) $0\checkmark$	(b) $-2/7 \pm \sqrt{96}/14i\checkmark$	(c) $1/32\checkmark$	(d) $42 - 15i\checkmark$	(e) $32/169\checkmark$	(f) $e^{i\pi/6}\checkmark$
θ	$iii : \mathbf{M}$	(a) $-4\checkmark$	(b) $1 \pm i\checkmark$	(c) $111/130\checkmark$	(d) $-6 - 2i\checkmark$	(e) $-5/4\checkmark$	(f) $729e^{i3\pi/2}\checkmark$
	$iv : \mathbf{E}$	(a) $8\checkmark$	(b) $3/2 \pm \sqrt{3}/2i\checkmark$	(c) $16/51\checkmark$	(d) $-17 - 6i\checkmark$	(e) $-16\checkmark$	(f) $e^{i16\pi/9}\checkmark$
	$i : \mathbf{T}$	(a) $-2\checkmark$	(b) $1 \pm \sqrt{48}/8i\checkmark$	(c) $23/60\checkmark$	(d) $19 - 33i\checkmark$	(e) $117/25\checkmark$	(f) $e^{i8\pi/15}\checkmark$
	$ii : \mathbf{R}$	(a) $-4\checkmark$	(b) $1/4 \pm \sqrt{23}/4i\checkmark$	(c) $5/3\checkmark$	(d) $20 + 15i\checkmark$	(e) $72/25\checkmark$	(f) $e^{i55\pi/28}\checkmark$
	$iii : \mathbf{I}$	(a) $-9\checkmark$	(b) $1/8 \pm \sqrt{47}/8i\checkmark$	(c) $-4/17\checkmark$	(d) $7 + 54i\checkmark$	(e) $-72/25\checkmark$	(f) $e^{i6\pi/7}\checkmark$
ι	$iv : \mathbf{P}$	(a) $-1\checkmark$	(b) $-5/18 \pm \sqrt{83}/18i\checkmark$	(c) $-11/4\checkmark$	(d) $1 + 15i\checkmark$	(e) $-1/9\checkmark$	(f) $343e^{i\pi/2}\checkmark$
	$i : \mathbf{I}$	(a) $-2\checkmark$	(b) $-1/14 \pm \sqrt{55}/14i\checkmark$	(c) $-17/13\checkmark$	(d) $-38 - 6i\checkmark$	(e) $-77/25\checkmark$	(f) $e^{i4\pi/9}\checkmark$
	$ii : \mathbf{D}$	(a) $-2\checkmark$	(b) $1/2 \pm \sqrt{51}/6i\checkmark$	(c) $-12/25\checkmark$	(d) $28 + 16i\checkmark$	(e) $117/25\checkmark$	(f) $-e^{i5\pi/3}\checkmark$
	$iii : \mathbf{E}$	(a) $6\checkmark$	(b) $-1/2 \pm \sqrt{32}/8i\checkmark$	(c) $7/6\checkmark$	(d) $40 - 15i\checkmark$	(e) $3\checkmark$	(f) $1\checkmark$
	$iv : \mathbf{A}$	(a) $-1\checkmark$	(b) $-1 \pm i\checkmark$	(c) $-3/2\checkmark$	(d) $12 + 20i\checkmark$	(e) $-72/25\checkmark$	(f) $125\checkmark$
κ	$i : \mathbf{F}$	(a) $-6\checkmark$	(b) $1/3 \pm \sqrt{44}/6i\checkmark$	(c) $1\checkmark$	(d) $-7 - 17i\checkmark$	(e) $63/25\checkmark$	(f) $1\checkmark$
	$ii : \mathbf{E}$	(a) $-2\checkmark$	(b) $3/2 \pm \sqrt{20}/4i\checkmark$	(c) $3/2\checkmark$	(d) $-4 - 32i\checkmark$	(e) $-33/25\checkmark$	(f) $e^{i2\pi/5}\checkmark$
	$iii : \mathbf{E}$	(a) $7\checkmark$	(b) $1/3 \pm \sqrt{8}/6i\checkmark$	(c) $-1/4\checkmark$	(d) $60 - 3i\checkmark$	(e) $3\checkmark$	(f) $e^{i5\pi/6}\checkmark$
	$iv : \mathbf{T}$	(a) $4\checkmark$	(b) $-3/8 \pm \sqrt{87}/8i\checkmark$	(c) $-7/12\checkmark$	(d) $-13 + 7i\checkmark$	(e) $16\checkmark$	(f) $e^{i\pi/2}\checkmark$
	$i : \mathbf{N}$	(a) $-1\checkmark$	(b) $1/4 \pm \sqrt{15}/4i\checkmark$	(c) $3/35\checkmark$	(d) $-27 + 42i\checkmark$	(e) $-24\checkmark$	(f) $e^{i13\pi/10}\checkmark$
μ	$ii : \mathbf{O}$	(a) $-9\checkmark$	(b) $3/16 \pm \sqrt{279}/16i\checkmark$	(c) $-31/25\checkmark$	(d) $-3 - 7i\checkmark$	(e) $63/169\checkmark$	(f) $e^{i11\pi/12}\checkmark$
	$iii : \mathbf{T}$	(a) $-1\checkmark$	(b) $5/4 \pm \sqrt{23}/4i\checkmark$	(c) $-3/10\checkmark$	(d) $-29 + 21i\checkmark$	(e) $63\checkmark$	(f) $1\checkmark$
	$iv : \mathbf{E}$	(a) $2\checkmark$	(b) $1 \pm \sqrt{24}/4i\checkmark$	(c) $3/2\checkmark$	(d) $4 + 33i\checkmark$	(e) $-1/4\checkmark$	(f) $e^{i2\pi/3}\checkmark$

Complex Numbers 1 (solutions)

ν	$i : \mathbf{S}$	(a) $-8\mathbf{X}$	(b) $-1/6 \pm \sqrt{20}/12i\checkmark$	(c) $3/2\checkmark$	(d) $39 + i\mathbf{X}$	(e) $143/25\mathbf{X}$	(f) $64e^{i4\pi/3}\checkmark$
	$ii : \mathbf{L}$	(a) $-5\checkmark$	(b) $-1/7 \pm \sqrt{108}/14i\checkmark$	(c) $15/34\checkmark$	(d) $17 + 58i\mathbf{X}$	(e) $-20\mathbf{X}$	(f) $e^{i\pi}\mathbf{X}$
	$iii : \mathbf{O}$	(a) $-9\checkmark$	(b) $-1/16 \pm \sqrt{31}/16i\mathbf{X}$	(c) $98/51\checkmark$	(d) $-36 - 26i\mathbf{X}$	(e) $5/4\checkmark$	(f) $25e^{i11\pi/6}\mathbf{X}$
	$iv : \mathbf{W}$	(a) $1\checkmark$	(b) $-1/4 \pm \sqrt{60}/8i\checkmark$	(c) $9/20\checkmark$	(d) $33 + 15i\checkmark$	(e) $187/25\checkmark$	(f) $e^{i2\pi/45}\mathbf{X}$
	$i : \mathbf{W}$	(a) $-3\checkmark$	(b) $3/8 \pm \sqrt{28}/16i\checkmark$	(c) $-27/130\checkmark$	(d) $-32 - 24i\checkmark$	(e) $-8\checkmark$	(f) $-e^{i7\pi/6}\mathbf{X}$
ξ	$ii : \mathbf{I}$	(a) $0\mathbf{X}$	(b) $1/2 \pm \sqrt{135}/10i\checkmark$	(c) $88/123\mathbf{X}$	(d) $11 - 23i\mathbf{X}$	(e) $72/25\mathbf{X}$	(f) $e^{i\pi}\checkmark$
	$iii : \mathbf{S}$	(a) $6\mathbf{X}$	(b) $7/10 \pm \sqrt{71}/10i\checkmark$	(c) $-17/5\checkmark$	(d) $27 - 16i\mathbf{X}$	(e) $-117/25\mathbf{X}$	(f) $e^{i29\pi/21}\checkmark$
	$iv : \mathbf{H}$	(a) $1\checkmark$	(b) $-1/2 \pm 1/2i\checkmark$	(c) $26/29\mathbf{X}$	(d) $47 + 48i\mathbf{X}$	(e) $-12\checkmark$	(f) $e^{i6\pi/5}\mathbf{X}$
	$i : \mathbf{C}$	(a) $9\checkmark$	(b) $1/4 \pm 1/4i\mathbf{X}$	(c) $1\mathbf{X}$	(d) $-13 - 39i\mathbf{X}$	(e) $-187/25\mathbf{X}$	(f) $e^{i40\pi/21}\checkmark$
	$ii : \mathbf{L}$	(a) $-2\checkmark$	(b) $-1/14 \pm \sqrt{195}/14i\checkmark$	(c) $19/15\checkmark$	(d) $19 + 42i\mathbf{X}$	(e) $-12/25\mathbf{X}$	(f) $-27e^{i6\pi/5}\mathbf{X}$
$\circ$	$iii : \mathbf{I}$	(a) $1\mathbf{X}$	(b) $-1/14 \pm \sqrt{111}/14i\checkmark$	(c) $21/25\mathbf{X}$	(d) $3 + 35i\mathbf{X}$	(e) $-35\mathbf{X}$	(f) $e^{i\pi}\checkmark$
	$iv : \mathbf{P}$	(a) $0\checkmark$	(b) $-5/14 \pm \sqrt{87}/14i\checkmark$	(c) $-58/51\checkmark$	(d) $-16 + 0i\mathbf{X}$	(e) $-1/4\mathbf{X}$	(f) $e^{i\pi/2}\checkmark$
	$i : \mathbf{P}$	(a) $-7\checkmark$	(b) $-1/3 \pm \sqrt{68}/6i\checkmark$	(c) $17/39\checkmark$	(d) $-9 + 21i\mathbf{X}$	(e) $187/25\mathbf{X}$	(f) $64\checkmark$
	$ii : \mathbf{L}$	(a) $1\checkmark$	(b) $1/6 \pm \sqrt{95}/6i\checkmark$	(c) $5/3\checkmark$	(d) $-43 - 48i\mathbf{X}$	(e) $12/25\mathbf{X}$	(f) $e^{i10\pi/7}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $10\checkmark$	(b) $3/2 \pm \sqrt{20}/4i\mathbf{X}$	(c) $97/123\mathbf{X}$	(d) $52 - 42i\mathbf{X}$	(e) $-5\mathbf{X}$	(f) $512e^{i3\pi/2}\mathbf{X}$
π	$iv : \mathbf{N}$	(a) $-6\checkmark$	(b) $1/6 \pm \sqrt{83}/6i\mathbf{X}$	(c) $-2/3\checkmark$	(d) $27 - 20i\mathbf{X}$	(e) $4\checkmark$	(f) $e^{i2\pi/3}\checkmark$
	$i : \mathbf{N}$	(a) $-11\checkmark$	(b) $1/3 \pm 2/3i\mathbf{X}$	(c) $1/8\checkmark$	(d) $36 + 40i\mathbf{X}$	(e) $-48\checkmark$	(f) $e^{i\pi/2}\checkmark$
	$ii : \mathbf{I}$	(a) $-1\mathbf{X}$	(b) $-1/2 \pm \sqrt{31}/2i\checkmark$	(c) $66/85\mathbf{X}$	(d) $-46 + 2i\mathbf{X}$	(e) $7/25\mathbf{X}$	(f) $729e^{i4\pi/3}\checkmark$
	$iii : \mathbf{N}$	(a) $12\checkmark$	(b) $4/5 \pm \sqrt{76}/10i\mathbf{X}$	(c) $-13/17\checkmark$	(d) $-79 - 129i\mathbf{X}$	(e) $8/25\checkmark$	(f) $256e^{i7/5}\checkmark$
	$iv : \mathbf{E}$	(a) $-3\checkmark$	(b) $-1/5 \pm 2/5i\mathbf{X}$	(c) $31/30\mathbf{X}$	(d) $-31 - 56i\mathbf{X}$	(e) $-1\checkmark$	(f) $e^{i3\pi/10}\mathbf{X}$
σ	$i : \mathbf{F}$	(a) $-7\checkmark$	(b) $-7/10 \pm \sqrt{11}/10i\checkmark$	(c) $-2/3\mathbf{X}$	(d) $46 + 13i\mathbf{X}$	(e) $20\mathbf{X}$	(f) $e^{i2\pi/3}\checkmark$
	$ii : \mathbf{A}$	(a) $1\checkmark$	(b) $1/6 \pm \sqrt{140}/12i\mathbf{X}$	(c) $-7/12\mathbf{X}$	(d) $-40 - 10i\mathbf{X}$	(e) $18\mathbf{X}$	(f) $e^{i6\pi/5}\mathbf{X}$
	$iii : \mathbf{R}$	(a) $1\checkmark$	(b) $1/10 \pm \sqrt{119}/10i\checkmark$	(c) $29/30\checkmark$	(d) $49 - 4i\mathbf{X}$	(e) $-33/169\checkmark$	(f) $4\mathbf{X}$
	$iv : \mathbf{M}$	(a) $0\checkmark$	(b) $-1/10 \pm \sqrt{19}/10i\mathbf{X}$	(c) $-50/35\checkmark$	(d) $5 - 5i\mathbf{X}$	(e) $-1\mathbf{X}$	(f) $-32e^{i11\pi/7}\checkmark$
	$i : \mathbf{W}$	(a) $0\checkmark$	(b) $5/6 \pm \sqrt{47}/6i\checkmark$	(c) $-81/82\checkmark$	(d) $18 - 62i\checkmark$	(e) $15/4\checkmark$	(f) $-8e^{i8\pi/5}\mathbf{X}$
τ	$ii : \mathbf{A}$	(a) $-1\checkmark$	(b) $1/6 \pm \sqrt{47}/6i\mathbf{X}$	(c) $2/29\mathbf{X}$	(d) $20 - 6i\mathbf{X}$	(e) $-54\mathbf{X}$	(f) $e^{i7\pi/6}\mathbf{X}$
	$iii : \mathbf{L}$	(a) $-15\checkmark$	(b) $1/6 \pm \sqrt{59}/6i\checkmark$	(c) $16/15\checkmark$	(d) $27 + 50i\mathbf{X}$	(e) $3\mathbf{X}$	(f) $e^{i5\pi/4}\mathbf{X}$
	$iv : \mathbf{L}$	(a) $4\checkmark$	(b) $1/6 \pm \sqrt{212}/12i\checkmark$	(c) $31/20\checkmark$	(d) $36 - 60i\mathbf{X}$	(e) $36\mathbf{X}$	(f) $9e^{i\pi/2}\mathbf{X}$
	$i : \mathbf{F}$	(a) $6\checkmark$	(b) $1/4 \pm \sqrt{7}/4i\checkmark$	(c) $-1/7\mathbf{X}$	(d) $-4 - 19i\mathbf{X}$	(e) $15\mathbf{X}$	(f) $-243e^{i\pi}\checkmark$
	$ii : \mathbf{O}$	(a) $-8\checkmark$	(b) $-1/3 \pm \sqrt{128}/12i\mathbf{X}$	(c) $-1/25\checkmark$	(d) $3 + 11i\mathbf{X}$	(e) $-32/25\checkmark$	(f) $e^{i2\pi/3}\mathbf{X}$
v	$iii : \mathbf{O}$	(a) $0\checkmark$	(b) $-3/8 \pm \sqrt{71}/8i\mathbf{X}$	(c) $-2\checkmark$	(d) $61 + 23i\mathbf{X}$	(e) $-12\checkmark$	(f) $e^{i3\pi/2}\mathbf{X}$
	$iv : \mathbf{D}$	(a) $-2\checkmark$	(b) $-1/6 \pm \sqrt{99}/18i\mathbf{X}$	(c) $0\mathbf{X}$	(d) $9 + 13i\mathbf{X}$	(e) $-45\checkmark$	(f) $81e^{i\pi}\checkmark$
	$i : \mathbf{S}$	(a) $-7\mathbf{X}$	(b) $-3/4 \pm \sqrt{31}/4i\checkmark$	(c) $37/17\checkmark$	(d) $-36 + 40i\mathbf{X}$	(e) $18\mathbf{X}$	(f) $1\checkmark$
	$ii : \mathbf{T}$	(a) $4\mathbf{X}$	(b) $-1 \pm \sqrt{48}/6i\checkmark$	(c) $-1/3\checkmark$	(d) $-9 + 74i\mathbf{X}$	(e) $-16\checkmark$	(f) $e^{i7\pi/4}\checkmark$
	$iii : \mathbf{A}$	(a) $-1\checkmark$	(b) $3/4 \pm \sqrt{7}/4i\mathbf{X}$	(c) $31/65\mathbf{X}$	(d) $-30 - i\mathbf{X}$	(e) $63/25\mathbf{X}$	(f) $e^{i19\pi/18}\mathbf{X}$
ϕ	$iv : \mathbf{R}$	(a) $7\checkmark$	(b) $3/2 \pm \sqrt{11}/2i\checkmark$	(c) $11/13\checkmark$	(d) $8 + 9i\mathbf{X}$	(e) $0\checkmark$	(f) $-512e^{i8\pi/5}\mathbf{X}$
	$i : \mathbf{C}$	(a) $1\checkmark$	(b) $3/16 \pm \sqrt{151}/16i\mathbf{X}$	(c) $-109/100\mathbf{X}$	(d) $15 + 20i\mathbf{X}$	(e) $12\mathbf{X}$	(f) $e^{i2\pi/7}\checkmark$
	$ii : \mathbf{L}$	(a) $10\checkmark$	(b) $3/8 \pm \sqrt{23}/8i\checkmark$	(c) $61/60\checkmark$	(d) $-56 - 16i\mathbf{X}$	(e) $-25\mathbf{X}$	(f) $e^{i7\pi/6}\mathbf{X}$
	$iii : \mathbf{U}$	(a) $2\checkmark$	(b) $1/10 \pm \sqrt{159}/10i\mathbf{X}$	(c) $6/41\checkmark$	(d) $-13 + 35i\checkmark$	(e) $12\mathbf{X}$	(f) $-e^{i11\pi/14}\mathbf{X}$
	$iv : \mathbf{B}$	(a) $0\checkmark$	(b) $-3/2 \pm \sqrt{19}/2i\checkmark$	(c) $3/5\mathbf{X}$	(d) $-41 - 14i\mathbf{X}$	(e) $117/25\mathbf{X}$	(f) $e^{i3\pi/2}\mathbf{X}$
χ	$i : \mathbf{W}$	(a) $3\checkmark$	(b) $5/16 \pm \sqrt{263}/16i\checkmark$	(c) $-2\checkmark$	(d) $27 - 39i\checkmark$	(e) $-77/25\checkmark$	(f) $e^{i2\pi/5}\mathbf{X}$
	$ii : \mathbf{I}$	(a) $-4\mathbf{X}$	(b) $5/6 \pm \sqrt{83}/6i\checkmark$	(c) $-17/32\mathbf{X}$	(d) $-34 - 28i\mathbf{X}$	(e) $119/169\mathbf{X}$	(f) $e^{i\pi/3}\checkmark$
	$iii : \mathbf{N}$	(a) $-1\checkmark$	(b) $-3/14 \pm \sqrt{47}/14i\mathbf{X}$	(c) $-1/8\checkmark$	(d) $18 - 16i\mathbf{X}$	(e) $-5/4\checkmark$	(f) $e^{i3\pi/2}\checkmark$
	$iv : \mathbf{G}$	(a) $-1\checkmark$	(b) $1/2 \pm \sqrt{12}/4i\checkmark$	(c) $8/21\mathbf{X}$	(d) $-22 - 14i\mathbf{X}$	(e) $-20\checkmark$	(f) $e^{i\pi/3}\checkmark$
	$i : \mathbf{C}$	(a) $-2\checkmark$	(b) $-1/3 \pm \sqrt{56}/6i\mathbf{X}$	(c) $1\mathbf{X}$	(d) $-67 - 16i\mathbf{X}$	(e) $36\mathbf{X}$	(f) $36\checkmark$
ω	$ii : \mathbf{H}$	(a) $1\checkmark$	(b) $1/3 \pm \sqrt{92}/6i\checkmark$	(c) $16/85\mathbf{X}$	(d) $-7 - 64i\mathbf{X}$	(e) $35\checkmark$	(f) $e^{i5\pi/4}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $0\checkmark$	(b) $1/4 \pm \sqrt{39}/4i\mathbf{X}$	(c) $-1/8\mathbf{X}$	(d) $4 - 15i\mathbf{X}$	(e) $-16\mathbf{X}$	(f) $e^{i2\pi/9}\mathbf{X}$
	$iv : \mathbf{T}$	(a) $4\mathbf{X}$	(b) $1/8 \pm \sqrt{28}/16i\checkmark$	(c) $-29/61\checkmark$	(d) $-5 + 23i\mathbf{X}$	(e) $0\checkmark$	(f) $e^{i3\pi/2}\checkmark$
	$i : \mathbf{C}$	(a) $-2\checkmark$	(b) $-1/3 \pm \sqrt{56}/6i\mathbf{X}$	(c) $1\mathbf{X}$	(d) $-67 - 16i\mathbf{X}$	(e) $36\mathbf{X}$	(f) $36\checkmark$
	$ii : \mathbf{H}$	(a) $1\checkmark$	(b) $1/3 \pm \sqrt{92}/6i\checkmark$	(c) $16/85\mathbf{X}$	(d) $-7 - 64i\mathbf{X}$	(e) $35\checkmark$	(f) $e^{i5\pi/4}\mathbf{X}$
$\square$	$iii : \mathbf{A}$	(a) $0\checkmark$	(b) $1/4 \pm \sqrt{39}/4i\mathbf{X}$	(c) $-1/8\mathbf{X}$	(d) $4 - 15i\mathbf{X}$	(e) $-16\mathbf{X}$	(f) $e^{i2\pi/9}\mathbf{X}$
	$iv : \mathbf{T}$	(a) $4\mathbf{X}$	(b) $1/8 \pm \sqrt{28}/16i\checkmark$	(c) $-29/61\checkmark$	(d) $-5 + 23i\mathbf{X}$	(e) $0\checkmark$	(f) $e^{i3\pi/2}\checkmark$
	$i : \mathbf{C}$	(a) $-2\checkmark$	(b) $-1/3 \pm \sqrt{56}/6i\mathbf{X}$	(c) $1\mathbf{X}$	(d) $-67 - 16i\mathbf{X}$	(e) $36\mathbf{X}$	(f) $36\checkmark$
	$ii : \mathbf{H}$	(a) $1\checkmark$	(b) $1/3 \pm \sqrt{92}/6i\checkmark$	(c) $16/85\mathbf{X}$	(d) $-7 - 64i\mathbf{X}$	(e) $35\checkmark$	(f) $e^{i5\pi/4}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $0\checkmark$	(b) $1/4 \pm \sqrt{39}/4i\mathbf{X}$	(c) $-1/8\mathbf{X}$	(d) $4 - 15i\mathbf{X}$	(e) $-16\mathbf{X}$	(f) $e^{i2\pi/9}\mathbf{X}$
$\square$	$iv : \mathbf{T}$	(a) $4\mathbf{X}$	(b) $1/8 \pm \sqrt{28}/16i\checkmark$	(c) $-29/61\checkmark$	(d) $-5 + 23i\mathbf{X}$	(e) $0\checkmark$	(f) $e^{i3\pi/2}\checkmark$
	$i : \mathbf{C}$	(a) $-2\checkmark$	(b) $-1/3 \pm \sqrt{56}/6i\mathbf{X}$	(c) $1\mathbf{X}$	(d) $-67 - 16i\mathbf{X}$	(e) $36\mathbf{X}$	(f) $36\checkmark$
	$ii : \mathbf{H}$	(a) $1\checkmark$	(b) $1/3 \pm \sqrt{92}/6i\checkmark$	(c) $16/85\mathbf{X}$	(d) $-7 - 64i\mathbf{X}$	(e) $35\checkmark$	(f) $e^{i5\pi/4}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $0\checkmark$	(b) $1/4 \pm \sqrt{39}/4i\mathbf{X}$	(c) $-1/8\mathbf{X}$	(d) $4 - 15i\mathbf{X}$	(e) $-16\mathbf{X}$	(f) $e^{i2\pi/9}\mathbf{X}$
	$iv : \mathbf{T}$	(a) $4\mathbf{X}$	(b) $1/8 \pm \sqrt{28}/16i\checkmark$	(c) $-29/61\checkmark$	(d) $-5 + 23i\mathbf{X}$	(e) $0\checkmark$	(f) $e^{i3\pi/2}\checkmark$